

Targeting Persistent Disagreement: Coarse Interventions in Social Networks*

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Abstract

Optimal targeting is determined not by who is most central or whose opinions are most extreme, but by which disagreement persists and who can move it. We study intervention in an opinion network when a decision-maker commits to one exposure rule and adjusts only its intensity. Reuse makes targeting an endogenous-path problem: the rule shapes the opinion path on which it will later be used and must therefore be chosen against the path it creates. Under equal centrality, uniform exposure is optimal exactly when updating erases inherited disagreement; otherwise exposure tilts toward disagreement the network carries forward. When centrality differs, exposure also follows agents' propagated leverage. The value of retargeting is governed by the effective dimension of the realized path: a fixed rule can match full targeting when it reproduces unrestricted intervention along that path. Combining Yale climate-opinion data with the Facebook friendship network, we find that California has substantially more authority than Texas but receives nearly identical exposure because Texas has greater discounted disagreement. In this application, the realized path remains nearly one-dimensional even after removing the common opinion component, so a single fixed rule captures almost all attainable gains.

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1 Introduction

Institutions that seek to shape public opinion redesign whom they reach less often than how intensely they intervene. Health agencies, political campaigns, and social-media platforms operate through exposure rules: which regions receive more advertising, which communities see more prominent messages, and which content is amplified or down-ranked in a feed. Such rules fix the relative pattern of attention; intensity adjusts more readily with budgets, enforcement effort, or ranking salience. Meta’s third-party fact-checking program is a case in point: introduced in 2016, it grew to nearly 100 organizations in more than 60 languages before Meta ended the U.S. program in 2025. Launching, expanding, or ending such a program required organizational, policy, and compliance decisions; the intensity with which flagged content was down-ranked or labelled could adjust more readily (Facebook, 2016; Meta, 2025b,a; Gorwa, 2019; Krishnan et al., 2021).

Institutions of this kind face a common design problem: the pattern of whom they reach changes slowly, while the strength of intervention can adjust more readily. We study this problem as *coarse targeting* in an opinion network. The decision-maker commits at the outset to an exposure vector $\mathbf{b} \in \mathbb{R}^n$ and then chooses a scalar intensity r_t in each period. With a single intervention date, the restriction is vacuous—any vector intervention $\mathbf{u}_0$ can be written as $\mathbf{b}r_0$. Its entire content is intertemporal: the same rule must be reused as opinions evolve.

Reuse makes the problem endogenous. The rule shapes the opinion path on which it will later be used, so it must be chosen against the path it creates. Two questions follow. What should a reusable rule target? And what is the value of targeting flexibility?

The first question is not answered by current extremity or centrality. Consider four agents divided into two symmetric islands, 1, 2 and 3, 4. Each agent places weight 0.4 on each opinion in her own island and 0.1 on each opinion in the other island, so all four agents have equal centrality. Suppose

$$\mathbf{x}_0 = (100, -99, 1, 1)'.$$

Agents 1 and 2 differ by 199, while the average opinions of the two islands differ by only 0.5. Social updating erases the within-island gap after one update, and optimal exposure therefore assigns agents 1 and 2 identical exposure, regardless of how far apart their opinions are. What survives is the much smaller disagreement between islands: the extreme opinions

in island 1 largely offset each other, while the common opinion in island 2 remains part of the path the network carries forward. Optimal targeting tilts toward island 2. A reusable rule targets disagreement according to whether the network preserves it, not according to its current magnitude.

The example isolates the first force: persistence. More generally, when agents are equally central – even in a directed, asymmetric network – uniform exposure is optimal exactly when updating eliminates inherited disagreement. If disagreement survives, exposure tilts toward it, and the response grows with its persistence. When centrality differs, a second force appears: leverage. Near a symmetric benchmark, a redistribution of authority propagates along network paths in Katz–Bonacich form, and optimal exposure tilts toward this effective influence rather than the raw shift. In the opinion-leader benchmark, greater authority attracts more exposure, but concentrating exposure on one leader creates temporary disagreement, so the optimal rule stays dispersed. Together, the two forces determine the value of exposure along the path the rule itself induces: persistence determines what remains worth moving, and leverage determines who can move it.

The second question is what targeting flexibility is worth. Our benchmark is full targeting, as in Li and Tan (2024), under which the decision-maker chooses an unrestricted intervention vector $\mathbf{u}_t$ in each period rather than requiring $\mathbf{u}_t = \mathbf{b}r_t$. We show that the value gap from committing to one rule is exactly the trace of a product of two matrices: one measures where the coarse and full policies differ, direction by direction; the other records how much discounted weight the endogenous opinion path places in each direction. A fixed rule can therefore attain the full-targeting value even when the two policies differ away from the realized path – equivalence holds exactly when the coarse rule reproduces the unrestricted intervention on the path that the rule itself induces. More generally, the gap is small when that path is concentrated in a few directions and the two policies agree on them. We call the number of directions carrying substantial path weight the *effective path dimension*. The value of targeting flexibility depends on the effective dimension of the realized path and on how closely the coarse and full policies agree along its leading directions, rather than directly on the number of agents or the apparent complexity of the network.

We illustrate both questions using U.S. state-level climate opinions and the Facebook Social Connectedness Index. The network supplies the propagation matrix, while the climate data provide initial disagreement across states. California has about 21% more authority weight

than Texas, yet receives only about 1% more exposure under the highest-value coarse rule we compute. Texas has greater discounted disagreement, which offsets most of California’s authority advantage. Ohio shows the complementary force: despite lower authority, its relatively high discounted disagreement raises its exposure above what authority alone would predict.

The application also shows why coarse targeting can perform well in a large asymmetric network. At the baseline, the leading direction carries 98.6% of discounted path weight; after removing the common opinion component, it still carries 96.2%, and the computed coarse rule captures 99.6% of attainable gains. Across the parameter grid, the leading share ranges from 95.6% to 99.8%. In this application, the realized path is therefore effectively one-dimensional, so a fixed forward-looking rule captures almost all gains from retargeting and outperforms a flexible policy that ignores future propagation.

The boundary is determined not by whether disagreement keeps arriving, but by how it changes effective path dimension. Recurrent agent-level shocks inject new disagreement in many unpredictable directions and can eventually make flexibility decisive. A persistent bias repeatedly injects the same source and, in the highly symmetric benchmark, adds at most one relevant disagreement direction. Recurrence alone does not defeat a reusable rule; diffuse new disagreement does.

Related literature

A. Network intervention and targeting. A large literature studies how outcomes in networks respond to targeted interventions. Early work emphasizes network position, centrality, and key-player measures (Galeotti and Goyal, 2009; Ballester et al., 2006); more recent work analyzes optimal intervention when agents interact strategically or spillovers are present (Candogan et al., 2022). Closest in method is Galeotti et al. (2020), who decompose a one-shot intervention into the principal components of a static network game. The spectral logic is related, but reuse changes the relevant object: persistence determines which opinion components remain valuable, and their weights are generated by the path the exposure rule itself induces rather than by a fixed network matrix alone. In dynamic opinion models, interventions are typically tailored across agents or adjusted as opinions evolve (Ghaderi and Srikant, 2014; Li and Tan, 2024). Li and Tan (2024) focuses on strategic competition among influencers with different agendas; here a single decision-maker commits to one

reusable exposure direction. Sadler (2023) studies how stochastic opinion fluctuations shape the concentration of influence campaigns in a network with stubborn agents. We instead study a decision-maker who chooses one exposure direction and reuses it over time. What matters is not centrality alone, but the leverage it provides¹

B. Dynamic systems and intervention. The paper is also related to work on intervention in dynamic networked systems. In engineering and applied mathematics, a large literature studies how intervention changes the evolution of network outcomes, often emphasizing feasibility, controllability, or stabilization (Lin, 1974; Pasqualetti et al., 2014; Liu and Barabási, 2016). Some recent work compares or selects where intervention enters the network (Summers et al., 2016; Lindmark and Altafini, 2018, 2020; Gao et al., 2021; Klickstein and Sorrentino, 2023).

Our problem differs in both the feasible instruments and the objective. The decision-maker chooses one exposure direction and reuses it over time rather than selecting arbitrary node-level interventions in each period. She also minimizes discounted losses along the entire opinion path rather than simply reaching or maintaining a desired outcome. We characterize the value of relaxing this restriction: it is determined by intervention error along the endogenous path, and hence by the path’s effective dimension, rather than merely by whether the system is controllable or stabilizable.

C. Opinion dynamics. We use linear updating as a tractable representation of social interaction (DeGroot, 1974; Friedkin and Johnsen, 1990; DeMarzo et al., 2003; Golub and Jackson, 2010). Our focus is not on whether opinions converge, but on how an institution should intervene when current exposure changes the future marginal value of influence. Models with behavioral or informational frictions, including locally Bayesian learning in Li and Tan (2020) and related work (Molavi et al., 2018; Li and Tan, 2021), often generate structured dynamics that can be approximated locally by linear updating rules. Our analysis applies to the local propagation structure of such richer environments.

D. Exposure design and platform governance. The paper also connects to work on exposure allocation in digital environments, where personalization is limited by privacy, regulation, transparency, or reputational concerns (Goldfarb and Tucker, 2011; Lambrecht and Tucker, 2013; Choi et al., 2019). These constraints motivate reusable exposure rules

¹Relatedly, Gaitonde et al. (2020) and Mostagir et al. (2022) examine how network structure shapes vulnerability to disagreement and misinformation.

rather than unrestricted personalization. We provide a dynamic theory of what such rules should target and when additional targeting flexibility is valuable.

2 Model

A network is represented by a weighted directed graph $G = (N, A)$, where $N = 1, \dots, n$ and $A = [a_{ij}]$ is row-stochastic, with $a_{ij} \geq 0$ and $\sum_j a_{ij} = 1$ for every i .² The entry a_{ij} is the weight agent i places on agent j 's opinion. A node may represent an individual, group, region, or other unit at which exposure is implemented. In Section 6, nodes are U.S. states.

Time is discrete. Let $\mathbf{x}_t \in \mathbb{R}^n$ denote opinions at date $t = 0, 1, 2, \dots$. After observing $\mathbf{x}_0$, a decision-maker commits at $t = 0$ to an exposure rule $\mathbf{b} \in \mathbb{R}^n$.³ In each period, she observes $\mathbf{x}_t$ and chooses a scalar intervention level $r_t \in \mathbb{R}$. Opinions evolve according to

$$\mathbf{x}_{t+1} = A\mathbf{x}_t + \mathbf{b}r_t.$$

The rule $\mathbf{b}$ fixes the relative exposure pattern across agents, while r_t scales and, when negative, reverses that pattern. The total intervention is $\mathbf{u}_t = \mathbf{b}r_t$.⁴

Exposure is measured relative to a no-intervention baseline. For a positive intensity, positive coordinates of $\mathbf{b}$ raise exposure above baseline and negative coordinates lower it. Thus $\mathbf{b}$ is a relative exposure rule, not an allocation of a fixed positive budget.⁵ The decision-maker cannot change this rule across agents over time. We call this restriction *coarse targeting*.

Coarse targeting restricts nothing within a single period: any vector intervention $\mathbf{u}_0$ can be written as $\mathbf{u}_0 = \mathbf{b}r_0$ with $\|\mathbf{b}\| = 1$. The restriction is entirely intertemporal – the same rule must be reused as opinions evolve.

We impose $\|\mathbf{b}\| \leq 1$, a capacity bound. Increasing $\|\mathbf{b}\|$ permits the same intervention path to be implemented with smaller scalar intensities and hence at lower cost. Lemma 2 shows

²We use boldface lowercase letters for vectors and boldface uppercase letters for matrices, and all vectors are column vectors.

³We maintain that A and $\mathbf{x}_0$ are known when the exposure rule is chosen. Bloch and Shabayek (2023) study targeting under uncertainty about network position.

⁴The linear updating rule can also be interpreted locally. For a smooth rule $\mathbf{x}_{t+1} = F(\mathbf{x}_t) + \mathbf{b}r_t$, deviations around a baseline $\bar{\mathbf{x}}$ satisfy $\Delta\mathbf{x}_{t+1} \approx J(\bar{\mathbf{x}})\Delta\mathbf{x}_t + \mathbf{b}r_t$. The relevant local propagation matrix is therefore $J(\bar{\mathbf{x}})$; see Appendix S3.

⁵Because $(\mathbf{b}, r_t)$ and $(-\mathbf{b}, -r_t)$ generate the same intervention, the rule is identified only up to sign.

that the bound is attained whenever intervention is strictly valuable.⁶

The technology applies whenever an institution runs one relative exposure rule rather than several independently adjustable rules in parallel.

Platform ranking. Nodes are communities or content categories, and $\mathbf{b}$ fixes their relative amplification or down-ranking from baseline. Changing the pattern requires product, policy, or compliance review, while its overall salience r_t adjusts more frequently.

Regulatory attention. Nodes are jurisdictions or firms, and $\mathbf{b}$ specifies how enforcement or outreach is reallocated from baseline. The pattern changes slowly, while the overall intensity of inspections, audits, or outreach responds to current conditions.

Fix $\mathbf{b}$, a discount factor $\delta \in (0, 1)$, and a cost parameter $c > 0$. The decision-maker chooses $\{r_t\}_{t \geq 0}$ to solve

$$\max_{\{r_t\}_{t \geq 0}} - \sum_{t=0}^{\infty} \delta^t (\mathbf{x}'_t Q \mathbf{x}_t + cr_t^2)$$

where $\mathbf{x}'_t Q \mathbf{x}_t$ is the loss from opinion deviations and cr_t^2 is the cost of scalar intervention. We normalize the decision-maker's common agenda to $\mathbf{0}$. This is without loss because $A\mathbf{1} = \mathbf{1}$: any common target can be removed by recentering opinions. The matrix Q is symmetric and positive definite and weights opinion deviations. Sections 3 and 5 require nothing further; Section 4 sets $Q = I$ so that losses do not mix eigendirections of A , and Section 6 takes Q diagonal with entries proportional to state population.

As a benchmark, under *full targeting* the decision-maker chooses an unrestricted intervention vector $\mathbf{u}_t \in \mathbb{R}^n$ in each period:

$$\mathbf{x}_{t+1} = A\mathbf{x}_t + \mathbf{u}_t,$$

with intervention cost $c|\mathbf{u}_t|^2$. When $\|\mathbf{b}\| = 1$, a coarse intervention $\mathbf{u}_t = \mathbf{b}r_t$ has the same cost, $c\|\mathbf{u}_t\|^2 = cr_t^2$. The two technologies therefore differ only in the ability to change the exposure pattern over time. Let $v^f(\mathbf{x}_0)$ denote the full-targeting value. Section 5 characterizes the value of this targeting flexibility.

For a fixed exposure rule $\mathbf{b}$, let $v(\mathbf{x}_0; \mathbf{b})$ denote the value obtained when intervention levels

⁶The Euclidean bound is invariant to orthonormal changes of basis. Other institutional restrictions, such as nonnegativity, may generate corner solutions.

are chosen optimally. The decision-maker chooses $\mathbf{b}$ to solve

$$\max_{\|\mathbf{b}\| \leq 1} v(\mathbf{x}_0; \mathbf{b}). \quad (1)$$

The exposure rule must be chosen against the opinion path that the rule itself induces. Sections 3 and 4 characterize what a reusable rule should target; Sections 5 and 7 characterize the value of targeting flexibility.

3 Characterizing optimal coarse targeting

We first formalize the claim in Section 2 that the coarse restriction is vacuous in a static problem. Suppose opinions evolve once according to $\mathbf{x}_1 = A\mathbf{x}_0 + \mathbf{b}r_0$ subject to $\|\mathbf{b}\| \leq 1$. The economically relevant object is the total intervention $\mathbf{u}_0 = \mathbf{b}r_0$. Any nonzero vector intervention $\mathbf{u}_0$ can be implemented, and its least costly representation has $\|\mathbf{b}\| = 1$; the zero intervention is immediate. The static problem is therefore equivalent to choosing $\mathbf{u}_0$ directly. If $QA\mathbf{x}_0 \neq \mathbf{0}$, the optimum is

$$\mathbf{u}_0^* = -\delta(cI + \delta Q)^{-1}QA\mathbf{x}_0.$$

Coarse targeting is therefore not a within-period restriction. It becomes substantive only because the same exposure rule must be reused.

For a fixed exposure rule $\mathbf{b} \neq \mathbf{0}$, the inner problem is standard. Its value is quadratic, so for a matrix $K^*(\mathbf{b})$ to be determined, write

$$v(\mathbf{x}; \mathbf{b}) = -\mathbf{x}'K^*(\mathbf{b})\mathbf{x}.$$

The Bellman equation is

$$v(\mathbf{x}; \mathbf{b}) = \max_r \left\{ -\mathbf{x}'Q\mathbf{x} - cr^2 - \delta(A\mathbf{x} + \mathbf{b}r)'K^*(\mathbf{b})(A\mathbf{x} + \mathbf{b}r) \right\}. \quad (2)$$

The optimal intervention level is linear,

$$r^*(\mathbf{x}; \mathbf{b}) = -\delta(c + \delta\mathbf{b}'K^*(\mathbf{b})\mathbf{b})^{-1}\mathbf{b}'K^*(\mathbf{b})A\mathbf{x} \equiv -\mathbf{p}(\mathbf{b})'\mathbf{x}, \quad (3)$$

and substitution gives the fixed-point equation

$$K^*(\mathbf{b}) = Q + \delta A' (K^*(\mathbf{b}) - \delta K^*(\mathbf{b}) \mathbf{b} (c + \delta \mathbf{b}' K^*(\mathbf{b}) \mathbf{b})^{-1} \mathbf{b}' K^*(\mathbf{b})) A, \quad (4)$$

capturing the discounted future payoff given the optimal intervention levels.

Lemma 1. *For every fixed $\mathbf{b} \neq \mathbf{0}$, $c > 0$, and $\delta \in (0, 1)$, there exists a unique symmetric positive definite matrix $K^*(\mathbf{b})$ solving (4). The optimal intervention level is given by (3).*

Under optimal intervention,

$$\mathbf{x}_{t+1} = \tilde{A}(\mathbf{b}) \mathbf{x}_t, \quad \tilde{A}(\mathbf{b}) \equiv A - \mathbf{b} \mathbf{p}(\mathbf{b})'.$$

The exposure rule therefore plays two roles: it determines how the decision-maker intervenes and the opinion path on which the same rule will later be used.

Lemma 2. *Suppose $Q A \mathbf{x}_0 \neq \mathbf{0}$. Then every optimal exposure rule satisfies $\|\mathbf{b}^*\| = 1$.*

Scaling up a fixed exposure direction implements the same intervention path with lower scalar-intervention cost. Since $Q A \mathbf{x}_0 \neq \mathbf{0}$ makes intervention strictly valuable, every optimum exhausts the capacity bound. The outer problem is therefore

$$\max_{\|\mathbf{b}\|=1} v(\mathbf{x}_0; \mathbf{b}) = -\mathbf{x}_0' K^*(\mathbf{b}) \mathbf{x}_0, \quad K^*(\mathbf{b}) \text{ solves (4)}. \quad (5)$$

The object linking the two roles of the exposure rule is the discounted opinion-path matrix

$$Z(\mathbf{b}) = \sum_{t=0}^{\infty} (\sqrt{\delta} \tilde{A}(\mathbf{b}))^t \mathbf{x}_0 \mathbf{x}_0' (\sqrt{\delta} \tilde{A}(\mathbf{b}))'^t. \quad (6)$$

It records where the path induced by $\mathbf{b}$ places its discounted weight: directions that are large or persistent receive greater weight. Standard perturbation results imply that $K^*(\mathbf{b})$, and hence $v(\mathbf{x}_0; \mathbf{b})$, vary smoothly with $\mathbf{b}$ and the primitives over the maintained parameter region.⁷

Proposition 1. *An optimal exposure rule exists. If $Q A \mathbf{x}_0 \neq \mathbf{0}$, any optimal exposure rule*

⁷See Konstantinov et al. (1993).

$\mathbf{b}^*$ satisfies, for some multiplier μ on $\|\mathbf{b}\|=1$,

$$\mu \mathbf{b}^* = K^*(\mathbf{b}^*) \tilde{A}(\mathbf{b}^*) Z(\mathbf{b}^*) \mathbf{p}(\mathbf{b}^*). \quad (7)$$

Proposition 1 captures the fixed point at the heart of coarse targeting.⁸ The exposure rule creates the path, and the path determines the marginal value of the rule; optimal exposure must therefore be chosen against the opinion path it creates. Persistence enters through $Z(\mathbf{b}^*)$: a direction that is large today but quickly disappears receives little weight, while a direction carried forward by the network enters losses repeatedly.⁹ The remaining terms in (7) translate those directions into the marginal value of reaching particular agents. Persistence determines what remains worth moving; leverage determines who can move it.

The characterization is homogeneous in initial opinions. Rescaling $\mathbf{x}_0$ by any nonzero scalar multiplies $Z(\mathbf{b})$ by its square and changes only the multiplier μ . It changes intervention intensity, not the optimal exposure direction. The same path matrix organizes the rest of the paper. Section 4 separates the two forces contained in (7), while Section 5 uses $Z(\mathbf{b})$ to characterize the value of targeting flexibility.

Because changing the exposure rule changes the path that gives the rule its value, the outer problem need not be concave and may have multiple stationary points. To conduct local comparative statics, we call a stationary point *nondegenerate* if the Hessian of the Lagrangian restricted to the tangent space of the unit sphere is nonsingular.

Corollary 1. *Suppose every stationary point of the outer problem (5) is nondegenerate. Then every stationary point is isolated and varies smoothly with the primitives; there are finitely many of them; and every local maximum is strict.*

For a scalar primitive θ , a local optimum satisfies

$$H(\mathbf{b}^*, \theta) \frac{d\mathbf{b}^*}{d\theta} = - (I - \mathbf{b}^* \mathbf{b}^{*'}) v_{\mathbf{b}\theta}(\mathbf{b}^*; \theta),$$

where $H(\mathbf{b}^*, \theta)$ is the constrained Hessian. Section 4 uses this relation to trace how optimal exposure responds to persistence and leverage.

⁸The characterization extends to substochastic matrices A satisfying $\rho(A) < 1$. Row stochasticity is used in Section 4 to separate average opinion from disagreement.

⁹The matrix $Z(\mathbf{b})$ solves a discrete Lyapunov equation, which we also exploit computationally.

4 Separating persistence from leverage

Persistent disagreement determines what remains worth moving; leverage determines who can move it. Two polar cases isolate the forces. Under equal centrality, persistent disagreement is the only reason to depart from uniform exposure. When inherited disagreement disappears after one update, unequal exposure instead reflects leverage. A joint local expansion then combines the two forces.

Throughout this section, let $Q = I$, and the uniform vector, average opinion, and initial disagreement be

$$\mathbf{u} = \mathbf{1}/\sqrt{n}, \quad \bar{x}_0 = \mathbf{1}'\mathbf{x}_0/n, \quad \mathbf{d} = \mathbf{x}_0 - \bar{x}_0\mathbf{1} = (I - \mathbf{u}\mathbf{u}')\mathbf{x}_0.$$

4.1 Persistent disagreement under equal centrality

Equal centrality switches off aggregate differences in agent leverage. In a row-stochastic network, agents are equally central when $\mathbf{1}'A = \mathbf{1}'$, or equivalently, when A is doubly stochastic. This condition allows directed and asymmetric interactions. Since $A\mathbf{x}_0 = \bar{x}_0\mathbf{1} + A\mathbf{d}$, the vector $A\mathbf{d}$ is precisely the inherited disagreement that survives one round of updating.

Proposition 2. *Suppose A is doubly stochastic and $Q = I$, and suppose $\bar{x}_0 \neq 0$. Then uniform targeting is optimal if and only if $A\mathbf{d} = \mathbf{0}$.*

Under equal centrality, persistent disagreement is not merely one reason for nonuniform targeting; it is the only reason. Uniform exposure moves average opinion but not disagreement.¹⁰ If $A\mathbf{d} = \mathbf{0}$, inherited disagreement disappears immediately, so nonuniform exposure only creates new dispersion. If $A\mathbf{d} \neq \mathbf{0}$, some disagreement survives and a nonuniform rule can move it.

The preceding characterization allows directed and asymmetric interactions. To obtain sharper comparative statics, the remainder of this subsection restricts attention to symmetric networks. Let

$$A = PDP', \quad D = \text{diag}(\lambda_1, \dots, \lambda_n), \quad \mathbf{p}_1 = \mathbf{u},$$

¹⁰The assumption $\bar{x}_0 \neq 0$ ensures that there is an average-opinion component for uniform exposure to target.

and let $\tilde{\mathbf{x}} = P'\mathbf{x}$ and $\tilde{\mathbf{b}} = P'\mathbf{b}$. The first coordinate is average opinion; each $j > 1$ is a disagreement dimension. The magnitude $|\lambda_j|$ measures how long that disagreement remains active, while its sign determines whether it maintains or reverses direction.¹¹

Lemma 3. *Suppose A is symmetric and $Q = I$. If $\lambda_j \tilde{x}_0^j = 0$ for some $j > 1$, then any strict local optimum satisfies $\tilde{b}_j^* = 0$.*

Because $A\mathbf{d} = \sum_{j>1} \lambda_j \tilde{x}_0^j \mathbf{p}_j$, Proposition 2 determines whether any disagreement remains relevant, while Lemma 3 rules out exposure to dormant dimensions. A highly symmetric family makes this geometry transparent. Consider

$$A(\rho) = \rho I + (1 - \rho) \frac{J}{n}, \quad \rho \in \left[-\frac{1}{n-1}, 1 \right].$$

Here, $A(\rho)\mathbf{u} = \mathbf{u}$ and $A(\rho)\mathbf{d} = \rho\mathbf{d}$, so every disagreement dimension has the same persistence ρ .

Corollary 2. *Suppose $Q = I$, $A = A(\rho)$, and $\mathbf{d} \neq \mathbf{0}$. Then any strict local maximum satisfies $\mathbf{b}^*(\rho) \in \text{span}\{\mathbf{u}, \mathbf{d}\}$.*

Let $\mathbf{b}^*(\rho) = \alpha\mathbf{u} + \beta \frac{\mathbf{d}}{\|\mathbf{d}\|}$. The induced dynamics reduce to

$$m_{t+1} = m_t + \alpha r_t, \quad y_{t+1} = \rho y_t + \beta r_t,$$

where $m_t = \mathbf{u}'\mathbf{x}_t$ and y_t is disagreement along $\mathbf{d}$. The fixed rule therefore divides its capacity between average opinion and $\mathbf{d}$, the single initially active direction of disagreement. At $\rho = 0$, disagreement disappears after one update and optimal exposure is uniform. At $\rho = 1$, no disagreement is removed, so optimal exposure is proportional to $\mathbf{x}_0$.

The two-island example in the introduction has the same structure. With n agents in each island, weight p_s on each same-island opinion, and weight p_d on each cross-island opinion, the group-divide dimension has eigenvalue $n(p_s - p_d)$, while every within-island dimension has eigenvalue zero. By Lemma 3, any strict local optimum ignores within-island disagreement, regardless of its magnitude, and assigns identical exposure within each island.

¹¹The change of basis rotates the state and exposure vector, not the instrument: the decision-maker still chooses one scalar intervention level r_t each period.

4.2 Agent leverage without persistent disagreement

Equal centrality isolated persistence by eliminating differences in agent leverage. We now isolate leverage by eliminating inherited disagreement. In a uniform opinion-leader network, absent intervention, all opinions collapse after one update. Unequal exposure therefore reflects differences in agents' leverage over future opinions.

Definition 1. A network is a **uniform opinion-leader network** if $A = \mathbf{1}\mathbf{a}'$, where $\mathbf{a}'\mathbf{1} = 1$ and $a_1 > a_2 > \dots > a_n > 0$.

Every agent listens to the same authority-weighted average $\mathbf{a}'\mathbf{x}$. Without intervention, $A\mathbf{x} = (\mathbf{a}'\mathbf{x})\mathbf{1}$, so the network carries no inherited disagreement beyond one update.

Proposition 3. Fix $\delta \in (0, 1)$ and $c > 0$, and suppose $Q = I$. For every initial opinion satisfying $\mathbf{a}'\mathbf{x}_0 \neq 0$, choose the sign normalization $\mathbf{a}'\mathbf{b}^* > 0$. The optimal exposure direction is independent of $\mathbf{x}_0$ and has the form

$$\mathbf{b}^* = \eta\mathbf{1} + \kappa\mathbf{a}, \quad \eta > 0, \quad \kappa > 0.$$

Consequently,

$$b_1^* > \dots > b_n^* > 0, \quad b_i^* - b_j^* = \kappa(a_i - a_j).$$

Exposure to high-authority agents moves the future consensus more, pulling the rule toward $\mathbf{a}$. But concentrated exposure creates temporary dispersion before the network averages opinions, pulling it back toward $\mathbf{1}$. The optimum balances the two: every agent stays active, $b_i^* > 0$, and authority determines the differences between them rather than which agents are reached. This holds even when a_1 is close to one, so that a single agent commands almost all authority. Optimal exposure thus targets authority while smoothing exposure across agents.

4.3 Persistence and leverage together

The polar cases isolate the two forces. We now turn both on around a common benchmark. Let $A_0 = PDP'$ be symmetric and row-stochastic, with $\mathbf{p}_1 = \mathbf{u}$. Using the homogeneity of the optimal exposure direction, normalize the benchmark average-opinion component to one and perturb opinions and the network according to

$$\tilde{\mathbf{x}}_0 = \mathbf{e}_1 + \boldsymbol{\xi}, \quad A_\varepsilon = A_0 + \varepsilon B, \quad B\mathbf{1} = \mathbf{0},$$

restricting attention to sufficiently small $\boldsymbol{\xi}$ and $|\varepsilon|$ such that A_ε remains nonnegative.

Define $\mathbf{s} = B'\mathbf{u}$. The component s_i is the first-order change in the aggregate weight placed on agent i 's opinion: $s_i > 0$ means that the perturbation increases this weight, while $s_i < 0$ means that it decreases it. Moreover, $\mathbf{u}'\mathbf{s} = 0$, so the perturbation redistributes influence across agents without changing its total.

At the benchmark, uniform exposure is optimal. Let $\tau \in (0, 1)$ denote the induced persistence of the average-opinion coordinate under the optimal intensity policy, so that

$$\tilde{x}_{t+1}^1 = \tau \tilde{x}_t^1.$$

Proposition 4. *Suppose $Q = I$ and $\boldsymbol{\xi} \perp \mathbf{e}_1$. There exist a strictly increasing function $M : [-1, 1] \rightarrow \mathbb{R}$ satisfying $M(0) = 0$, a scalar $\tau \in (0, 1)$, and coefficients $\chi_j > 0$, depending only on (A_0, δ, c) , such that, defining*

$$\mathbf{g} = (I - \delta\tau A'_0)^{-1}\mathbf{s} = \sum_{j>1} g_j \mathbf{p}_j,$$

the unique strict local optimum near $\mathbf{e}_1$ satisfies

$$\tilde{\mathbf{b}}^*(\boldsymbol{\xi}, \varepsilon) = \mathbf{e}_1 + \sum_{j>1} [M(\lambda_j)\xi_j + \varepsilon\chi_j g_j] \mathbf{e}_j + o(\|\boldsymbol{\xi}\| + |\varepsilon|).$$

Setting $\varepsilon = 0$ isolates persistence. The coefficient $M(\lambda_j)$ is the exposure response per unit of initial disagreement in dimension j . It balances the discounted benefit of moving that disagreement against the cost of diverting exposure from average opinion. Since $M(0) = 0$, a dimension eliminated after one update receives no first-order response. The response has the sign of λ_j : exposure tilts toward current disagreement when $\lambda_j > 0$ and against it when $\lambda_j < 0$. Because M is strictly increasing, the tilt strengthens with λ_j .

Setting $\boldsymbol{\xi} = \mathbf{0}$ isolates leverage. The decision-maker responds not to the direct redistribution $\mathbf{s}$, but to its propagated value

$$\mathbf{g} = \mathbf{s} + \delta\tau A'_0 \mathbf{s} + \delta^2 \tau^2 (A'_0)^2 \mathbf{s} + \dots$$

The factor $\delta\tau$ reflects both discounting and the decline of the benchmark average-opinion

state under optimal intervention. Thus $\mathbf{g}$ is a signed Katz–Bonacich measure of effective leverage. A change in an agent’s direct influence matters more when it is transmitted through influential network paths. Network position converts raw authority into leverage over future opinions.

The clique connects this local response to Proposition 3. Let

$$A_0 = \frac{J}{n}, \quad B = \mathbf{1} \left(\mathbf{a} - \frac{1}{n} \mathbf{1} \right)',$$

so that

$$A_\varepsilon = \mathbf{1} \mathbf{a}'_\varepsilon, \quad \mathbf{a}_\varepsilon = (1 - \varepsilon) \frac{1}{n} \mathbf{1} + \varepsilon \mathbf{a}.$$

Here

$$\mathbf{s} = \sqrt{n} \left(\mathbf{a} - \frac{1}{n} \mathbf{1} \right),$$

and $A'_0 \mathbf{s} = \mathbf{0}$, so effective leverage equals the direct authority shift. Symmetry also makes the coefficients χ_j identical across disagreement dimensions. Returning to the original coordinates, Proposition 4 therefore implies

$$\mathbf{b}^*(\varepsilon) = \mathbf{u} + \varepsilon C \left(\mathbf{a} - \frac{1}{n} \mathbf{1} \right) + o(|\varepsilon|), \quad C > 0.$$

Thus $b_i^* - b_j^*$ is proportional to $a_i - a_j$. This is the local counterpart of Proposition 3, showing the same pairwise exposure pattern holds globally throughout the opinion-leader family.

Persistence and leverage therefore add to first order. They can be isolated because they are the two margins of a single local response; their interaction enters only at higher order.

5 The value of targeting flexibility

The same endogenous path that determines the optimal coarse rule also determines the value of relaxing it. Recall the full-targeting benchmark from Section 2, under which the decision-maker can choose an unrestricted intervention vector each period. Full flexibility is valuable only when the path induced by a coarse rule visits directions in which one fixed exposure rule cannot reproduce the full intervention.

Let $v^f(\mathbf{x}_0)$ denote the full-targeting value. Since every coarse-targeting path is feasible

under full targeting,

$$v^f(\mathbf{x}_0) \geq v(\mathbf{x}_0; \mathbf{b}).$$

Define

$$\Delta(\mathbf{b}) = v^f(\mathbf{x}_0) - v(\mathbf{x}_0; \mathbf{b}) \geq 0, \quad \Delta^* = \min_{\|\mathbf{b}\|=1} \Delta(\mathbf{b}).$$

To characterize the gap, compare the intervention matrices. Write the full-targeting value as

$$v^f(\mathbf{x}) = -\mathbf{x}' K^f \mathbf{x}.$$

Under full targeting,

$$\mathbf{u}_t = -F^f \mathbf{x}_t, \quad F^f = \delta(cI + \delta K^f)^{-1} K^f A.$$

Under coarse targeting,

$$\mathbf{u}_t = -F^c(\mathbf{b}) \mathbf{x}_t, \quad F^c(\mathbf{b}) = \mathbf{b} \mathbf{p}(\mathbf{b})'.$$

Define

$$\Phi(\mathbf{b}) = (F^c(\mathbf{b}) - F^f)'(cI + \delta K^f)(F^c(\mathbf{b}) - F^f).$$

This weighting matrix captures both the current cost of intervention error and its continuation cost through next-period opinions. The total loss accumulates this intervention error along the path generated by the coarse rule. Recall that $Z(\mathbf{b})$ is the discounted path matrix defined in (6).

Proposition 5. *For any exposure rule satisfying $\|\mathbf{b}\|=1$,*

$$\Delta(\mathbf{b}) = \sum_{t=0}^{\infty} \delta^t \mathbf{x}_t' \Phi(\mathbf{b}) \mathbf{x}_t = \text{tr}(\Phi(\mathbf{b}) Z(\mathbf{b})),$$

where $\mathbf{x}_{t+1} = \tilde{A}(\mathbf{b}) \mathbf{x}_t$, $t \geq 0$.

A fixed one-dimensional rule can attain the full-targeting value even when its policy differs from the full policy away from the realized path. Since $cI + \delta K^f$ is positive definite,

$$\Delta(\mathbf{b}) = 0 \iff (F^c(\mathbf{b}) - F^f) \mathbf{z} = \mathbf{0} \text{ for every } \mathbf{z} \in \text{Range}(Z(\mathbf{b})).$$

Thus the coarse rule need not reproduce full targeting for every possible opinion vector. It need only reproduce it on the path that the rule itself induces. Because $Z(\mathbf{b})$ depends on $\mathbf{b}$, this is an endogenous fixed-point condition, not an approximation against a fixed path.

To measure approximate performance, write

$$Z(\mathbf{b}) = \sum_{k=1}^n \sigma_k \mathbf{q}_k \mathbf{q}_k', \quad \sigma_1 \geq \cdots \geq \sigma_n \geq 0.$$

Then

$$\Delta(\mathbf{b}) = \sum_{k=1}^n \sigma_k \mathbf{q}_k' \Phi(\mathbf{b}) \mathbf{q}_k.$$

Thus the gap is small when discounted path weight is concentrated in a few directions and the coarse and full policies agree closely on those directions. A convenient scalar summary of path concentration is the effective path dimension

$$d_{\text{eff}}(Z(\mathbf{b})) = \frac{\text{tr}(Z(\mathbf{b}))^2}{\text{tr}(Z(\mathbf{b})^2)}. \quad (8)$$

It equals one when the path is rank one and increases as discounted path weight spreads across directions. The relevant dimension is therefore not the number of agents, but the number of directions carrying substantial path weight.

Effective dimension describes approximate performance with one reusable rule. A related exact question is how many independently adjustable but reusable exposure rules are needed to recover full targeting. Suppose the decision-maker can choose each intervention from a fixed k -dimensional subspace. Let

$$Z^f = \sum_{t=0}^{\infty} \delta^t \mathbf{x}_t^f \mathbf{x}_t^{f'}$$

denote the discounted opinion-path matrix under full targeting. Such a technology implements full targeting from $\mathbf{x}_0$ if and only if its exposure subspace contains every intervention $F^f \mathbf{x}_t^f$ used along that path. Hence the minimum required targeting dimension is

$$k_{\min} = \text{rank}(F^f Z^f F^{f'}).$$

Exact implementation may require many exposure directions even when most attainable

gains can be captured with one. This distinction is why approximate path concentration, rather than exact rank alone, is the relevant welfare diagnostic.

The preceding diagnostics depend on the realized targeting path. For symmetric networks, persistence also yields a bound using only primitives.

Corollary 3. *Suppose A is symmetric and $Q = I$. Let $\mathbf{d}$ denote initial disagreement and let $|\lambda_2|$ be the largest absolute eigenvalue of A on the disagreement space. Then*

$$0 \leq \Delta^* \leq \frac{\delta |\lambda_2|^2}{1 - \delta |\lambda_2|^2} \|\mathbf{d}\|^2.$$

To see the bound, consider the feasible uniform rule. Intervention then affects only average opinion, while disagreement evolves as $A^t \mathbf{d}$. Hence

$$\Delta^* \leq \Delta(\mathbf{u}) \leq \sum_{t=1}^{\infty} \delta^t \|A^t \mathbf{d}\|^2 \leq \frac{\delta |\lambda_2|^2}{1 - \delta |\lambda_2|^2} \|\mathbf{d}\|^2.$$

This primitive bound on the welfare gap is distinct from a bound on effective path dimension. Fast elimination of disagreement can limit the value of flexibility even when the realized path is not itself rank one; path concentration also depends on initial opinions and the induced policy.

The theory thus identifies two measurable diagnostics: how concentrated the realized path is, and how accurately the coarse rule acts on its leading directions. Section 6 evaluates both.

6 Empirical illustration: climate-opinion disagreement

We illustrate the theory using U.S. state-level climate opinions and the Facebook Social Connectedness Index (SCI). We do not estimate opinion dynamics. We use observed disagreement and social connectedness to compute the exposure rule and the value of targeting flexibility implied by the model.

6.1 Data and construction

We use state-level climate opinions from the 2023 edition of the Yale Climate Opinion Maps (Howe et al., 2015; Marlon et al., 2022). The initial opinion $x_{0,s}$ is the estimated share of

residents in state s who disagree with the statement “Most scientists think global warming is happening.” We interpret this variable as disagreement with the scientific-consensus benchmark and normalize the decision-maker’s agenda to zero.

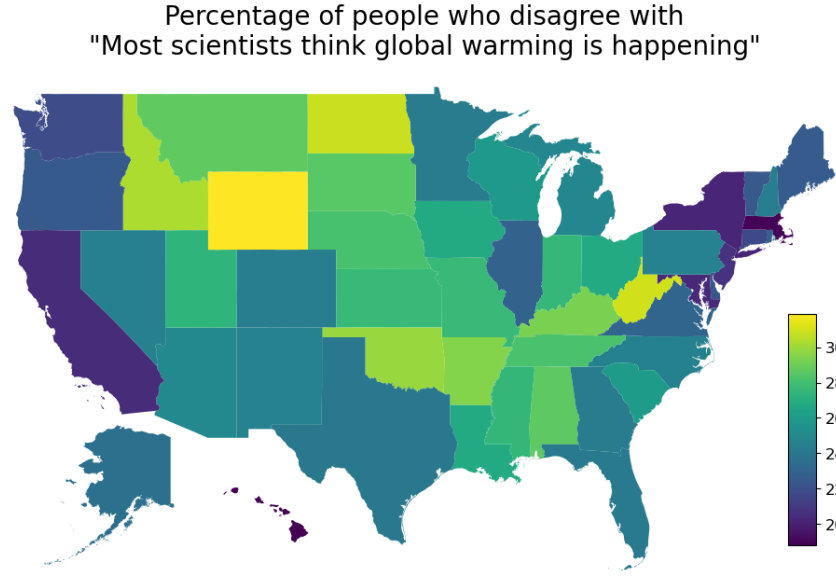


Figure 1: Disagreement with the scientific consensus across U.S. states.

We measure network links using the Facebook SCI, aggregated from counties to states following Bailey et al. (2021). The construction derived in Appendix S5 gives

$$\hat{A}_{ss'} = \frac{\text{SCI}_{ss'} N_{s'}}{\sum_{s''} \text{SCI}_{ss''} N_{s''}},$$

where $N_{s'}$ is the population of destination state s' . Thus $\hat{A}_{ss'}$ is the weight state s places on opinions in state s' . The resulting row-stochastic network is asymmetric and geographically clustered, as shown in Figure 2. Authority and persistent disagreement therefore need not point toward the same states: a state may have high leverage over future opinions, or it may be connected to disagreement that the network continues to carry forward.

6.2 Which states receive exposure?

We compute the highest-value stationary coarse rule found by the multi-start procedure described in Appendix S5.2, and denote it by $\mathbf{b}^*$. Our baseline uses $\delta = 0.9$, $c = 10$, and

Ohio provides the complementary comparison. Its discounted-disagreement score is higher than those of both Texas and California, while its authority weight is much lower. It therefore receives less exposure in levels, but substantially more than authority alone would suggest: its exposure weight is 0.220, compared with an authority weight of 0.160. Thus persistent disagreement offsets an authority advantage in the Texas–California comparison and amplifies exposure relative to authority in Ohio. Table 1 reports these comparisons.

Table 1: Authority, discounted disagreement, and exposure ($\delta = 0.9$, $c = 10$). Authority and exposure directions have unit Euclidean norm.

	Texas	California	Ohio
Authority centrality	0.343	0.414	0.160
Discounted disagreement score	−0.078	−0.182	−0.004
Exposure, $Q = Q^{\text{pop}}$	0.325	0.328	0.220

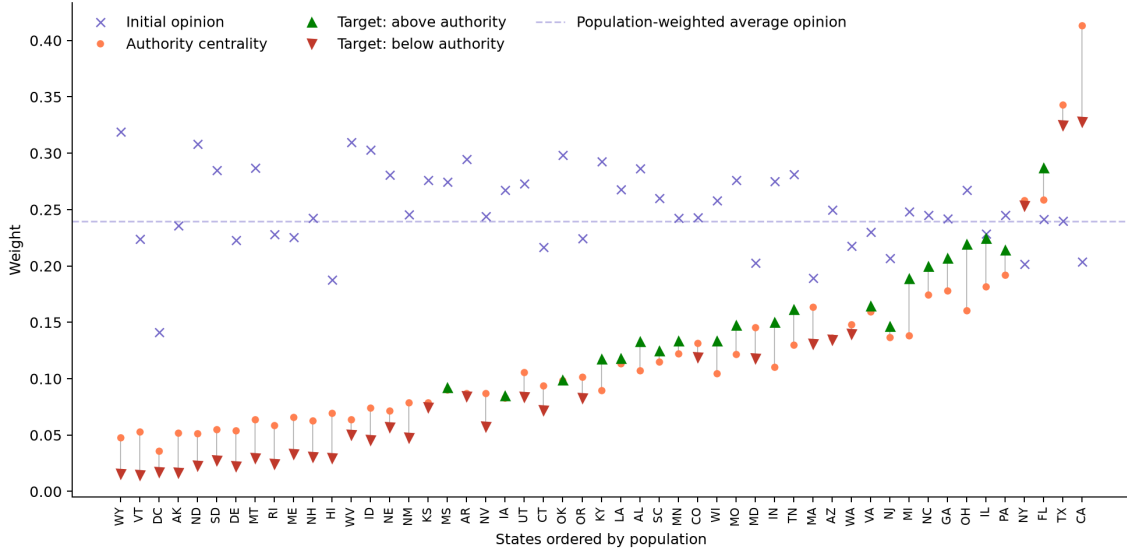


Figure 3: Population-weighted exposure compared with authority centrality and the initial opinion across states.

6.3 How valuable is targeting flexibility?

We compare the computed coarse rule with full targeting, several fixed directions, and a myopic vector policy. Fixed directions use their optimal dynamic scalar policies, while the

myopic rule reoptimizes an unrestricted one-period vector intervention each period. For a rule with value v , we report

$$\frac{v - v^{\emptyset}}{v^f - v^{\emptyset}},$$

the share of attainable gains captured by the rule. The measure is one under full targeting and zero under no intervention.

Table 2: Share of attainable gains captured under population-weighted payoff losses ($Q = Q^{\text{pop}}$), rounded to the nearest tenth of a percentage point.

	$c = 1$	$c = 10$	$c = 100$	$c = 1000$	$c = 10000$
Opinion ($\mathbf{b} \propto \mathbf{x}_0$)	98.3%	91.1%	72.0%	51.3%	45.2%
Network-updated ($\mathbf{b} \propto \hat{A}\mathbf{x}_0$)	98.5%	91.7%	73.3%	53.0%	46.9%
Payoff-updated ($\mathbf{b} \propto Q^{\text{pop}}\hat{A}\mathbf{x}_0$)	94.3%	97.8%	99.4%	99.4%	99.3%
Authority ($\mathbf{b} \propto \mathbf{v}_1$)	97.2%	99.1%	97.8%	94.2%	92.6%
Static direction, dynamic intensity	99.6%	99.2%	99.5%	99.4%	99.3%
Period-by-period myopic retargeting	99.7%	91.5%	52.2%	25.1%	20.2%
Coarse ($\mathbf{b} = \mathbf{b}^*$)	99.9%	99.9%	100.0%	100.0%	100.0%

Across the displayed grid, the computed coarse rule captures about 99.9% or more of attainable gains. Thus commitment changes whom the decision-maker targets, as the preceding comparisons show, but once the rule is chosen against the path it creates, subsequent retargeting adds little value in this deterministic network. Careful ex ante choice of the exposure rule can therefore substitute for ex post flexibility.

The myopic vector policy can retarget freely every period, yet its capture falls from 99.7% at $c = 1$ to 20.2% at $c = 10000$. Flexibility does not compensate for ignoring continuation values: here a fixed forward-looking rule substantially outperforms an unrestricted one-period policy.

The fixed heuristics isolate different pieces of the targeting problem. The exact one-period direction is

$$\mathbf{b}^{\text{static}} \propto (cI + \delta Q^{\text{pop}})^{-1} Q^{\text{pop}} \hat{A}\mathbf{x}_0,$$

and therefore incorporates current opinions, one-step propagation, payoff weights, and intervention cost, but not the future reuse of the rule. Its loss remains below 0.9% throughout the grid. By contrast, the opinion and network-updated rules, $\mathbf{b} \propto \mathbf{x}_0$ and $\mathbf{b} \propto \hat{A}\mathbf{x}_0$, ignore payoff weights and deteriorate sharply as c rises. Authority performs best near the baseline

but less well away from it.

The trace identity in Proposition 5 explains why the value of retargeting is small:

$$\Delta(\mathbf{b}^*) = \text{tr}(\Phi(\mathbf{b}^*)Z(\mathbf{b}^*)).$$

At the baseline, the realized path has effective dimension 1.03, and its leading direction carries 98.59% of discounted path weight. The coarse and full policies agree especially closely on the leading path direction. Thus the policy disagreement is particularly small on the direction that matters most.¹²

The baseline initial state is dominated by its common component, which accounts for 98.0% of $\|\mathbf{x}_0\|^2$. Hence $\mathbf{x}_0$ points close to the average-opinion direction. By scale invariance, the issue is not the magnitude of $\mathbf{d}$: rescaling it would not change the optimal exposure direction or the normalized path diagnostics. The relevant question is whether the concentration result survives when the initial state lies entirely in the disagreement space. We therefore replace $\mathbf{x}_0$ by $\mathbf{d}$ and recompute the coarse and full policies, holding $\hat{A}$, Q^{pop} , δ , and c fixed. This is a counterfactual diagnostic that isolates the observed disagreement direction, not a replacement for the baseline specification.

Table 3: Baseline and centered-disagreement paths ($\delta = 0.9$, $c = 10$, $Q = Q^{\text{pop}}$).

Initial state	Coarse capture	$d_{\text{eff}}(Z)$	Lead share
$\mathbf{x}_0$	99.91%	1.03	98.59%
$\mathbf{d}$	99.61%	1.08	96.24%

Starting from centered cross-state disagreement makes the realized path somewhat less concentrated: effective dimension rises from 1.03 to 1.08, and the leading-direction share falls from 98.59% to 96.24%. The change is modest, however, and the computed coarse rule still captures 99.61% of attainable gains. Thus the concentration result is not confined to an initial state close to the average-opinion direction: the observed disagreement direction and the Facebook network also generate a highly concentrated path.

Recurrent shocks provide the boundary. The observed cross-state standard deviation of initial disagreement is about 0.036, so $\sigma = 0.1$ is already almost three times the empirical

¹²The same pattern holds across the population-weighted parameter grid. Effective path dimension never exceeds 1.09, and the leading direction carries between 95.6% and 99.8% of discounted path weight.

dispersion. At the baseline parameters, coarse targeting captures 97.2% of attainable gains when $\sigma = 0.1$, but capture falls to 78.2% at $\sigma = 0.5$ and 72.4% at $\sigma = 1$. A fixed rule performs well when future disagreement follows a concentrated and predictable path; retargeting becomes valuable when recurrent shocks continually populate new directions (Appendix Table 5).

7 The dimension of new disagreement

Section 5 showed that the value of targeting flexibility depends on the effective dimension of the opinion path. We now ask what raises that dimension. Recurrent agent-level shocks inject disagreement in many unpredictable directions and can overwhelm any fixed exposure rule as the number of directions grows. A persistent bias instead injects the same source repeatedly, generating a predictable propagated subspace. In the highly symmetric benchmark, it adds at most one relevant disagreement direction.

7.1 Recurrent shocks and dimensional flexibility

Recurrent agent-level shocks raise effective path dimension because each realization injects disagreement in a new direction. Suppose shocks are iid and arrive after intervention:

$$\mathbf{x}_{t+1} = A\mathbf{x}_t + \mathbf{u}_t + \boldsymbol{\nu}_t, \quad \boldsymbol{\nu}_t \sim \mathcal{N}(\mathbf{0}, \sigma^2 I).$$

The decision-maker observes $\mathbf{x}_t$ before choosing $\mathbf{u}_t$. We restrict attention to symmetric networks and write

$$A = PDP', \quad \tilde{\mathbf{x}}_t = P'\mathbf{x}_t, \quad \tilde{\boldsymbol{\nu}}_t = P'\boldsymbol{\nu}_t.$$

Since P is orthonormal, $\tilde{\boldsymbol{\nu}}_t \sim \mathcal{N}(\mathbf{0}, \sigma^2 I)$.

We compare three rules under the same intervention-cost normalization. Under *full targeting*, the decision-maker chooses an unrestricted vector $\tilde{\mathbf{u}}_t$. Under *coarse targeting*, intervention is restricted to $\tilde{\mathbf{u}}_t = \tilde{\mathbf{b}}r_t$, with $\|\tilde{\mathbf{b}}\| = 1$ fixed at $t = 0$. Under *myopic targeting*, the decision-maker retains the unrestricted vector technology but treats the next period as terminal:

$$\tilde{\mathbf{u}}_t^m \in \arg \min_{\tilde{\mathbf{u}}} \left\{ c\|\tilde{\mathbf{u}}\|^2 + \delta\|D\tilde{\mathbf{x}}_t + \tilde{\mathbf{u}}\|^2 \right\}.$$

Thus

$$\tilde{\mathbf{u}}_t^m = -\frac{\delta}{c + \delta} D \tilde{\mathbf{x}}_t.$$

Myopic targeting retains dimensional flexibility but ignores the persistence of the disagreement it moves.

Because the shocks are additive and have mean zero, the optimal response to current opinions is unchanged. For each rule $q \in \{f, m, c\}$, the value becomes

$$\tilde{v}^q(\tilde{\mathbf{x}}_0) = -\tilde{\mathbf{x}}_0' \tilde{K}^q \tilde{\mathbf{x}}_0 - \frac{\delta \sigma^2}{1 - \delta} \text{tr}(\tilde{K}^q).$$

The first term is the loss generated by initial opinions; the second is the discounted loss generated by recurrent shocks.

Proposition 6. *Suppose A is symmetric, $\boldsymbol{\nu}_t \sim \mathcal{N}(\mathbf{0}, \sigma^2 I)$, and $\sigma > 0$.*

(1) *Full targeting strictly dominates myopic targeting. Moreover, there exist constants $0 < \underline{\eta} \leq \bar{\eta}$, depending only on the primitives, such that*

$$\frac{1}{1 + \bar{\eta}} \leq \frac{\tilde{v}^f(\tilde{\mathbf{x}}_0)}{\tilde{v}^m(\tilde{\mathbf{x}}_0)} \leq \frac{1}{1 + \underline{\eta}}.$$

(2) *Full targeting weakly dominates coarse targeting, with equality if $\text{rank}(A) = 1$ and strict dominance if $\text{rank}(A) > 1$.*

The proposition separates two advantages of full targeting. Relative to myopic targeting, it adds foresight while holding dimensional flexibility fixed. Relative to coarse targeting, it adds dimensional flexibility while holding foresight fixed. The substantive question is which advantage matters more.

Recall the highly symmetric family

$$A(\rho) = \rho I + (1 - \rho) \frac{J}{n}, \quad \rho \in (0, 1).$$

Proposition 7. *Suppose $\sigma > 0$ and $\mathbf{d} \neq \mathbf{0}$.*

(1) For each fixed n , an optimal coarse exposure rule can be chosen in

$$\text{span}\{\mathbf{u}, \mathbf{d}\}.$$

(2) Consider a sequence of such networks as $n \rightarrow \infty$. If

$$|\mathbf{u}'_n \mathbf{x}_0^{(n)}| = O(1) \quad \text{and} \quad \|\mathbf{d}_n\| = O(1),$$

then

$$1 > \lim_{n \rightarrow \infty} \frac{\tilde{v}_n^f}{\tilde{v}_n^m} > \lim_{n \rightarrow \infty} \frac{\tilde{v}_n^f}{\tilde{v}_n^c} > 0.$$

Part (1) shows that the span restriction from Section 4.1 survives recurrent shocks. Isotropy makes all disagreement directions orthogonal to $\mathbf{d}$ ex ante symmetric, so an optimal fixed rule need not select among them. The shocks nevertheless realize disagreement in every one of those directions.

Part (2) delivers a reversal. Myopic targeting ignores persistence but can react separately to every realized direction. Coarse targeting is forward-looking but confined to one fixed exposure pattern. As n grows, the losses generated in directions omitted by that pattern dominate, so dimensional flexibility becomes more valuable than foresight.

This is an asymptotic stochastic boundary, not the ranking in every finite network. In the deterministic finite network studied in Section 6, foresight dominates and coarse targeting substantially outperforms myopic targeting.

The polar cases make the mechanism transparent. In the clique ($\rho = 0$), disagreement disappears after one update, so coarse targeting matches full targeting. In the identity network, the deterministic path from a fixed initial state remains concentrated, but recurrent shocks populate every disagreement direction. The value of a fixed exposure rule therefore falls relative to full targeting as σ increases (Figure 4).

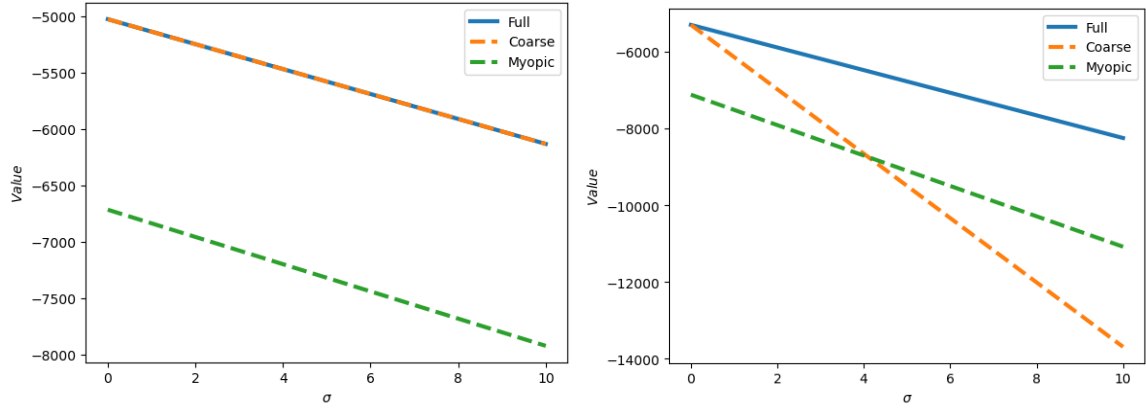


Figure 4: Value comparison across targeting methods in the clique (left) and the identity network (right) as the scale of recurrent shocks increases.

7.2 Persistent bias and predictable disagreement

Recurrent shocks raise effective dimension because each realization arrives in a new direction. A persistent bias repeatedly injects the same source. Consider

$$\mathbf{x}_{t+1} = A\mathbf{x}_t + \mathbf{b}r_t + \mathbf{w},$$

where $\mathbf{w}$ is time invariant.¹³ For any fixed exposure rule, the optimal intervention is affine in current opinions.

Because the same bias is injected every period, its effect is repeatedly propagated through the network: $\mathbf{w}$, $A\mathbf{w}$, $A^2\mathbf{w}$ In a general network, these successive effects may point in several different directions. Unlike recurrent idiosyncratic shocks, however, all of these directions are determined in advance. Proposition S8 in the appendix shows that optimal exposure need only respond to these predictable effects and to inherited disagreement that survives social updating.

In the highly symmetric family, the dimensional implication is sharp.

Corollary 4. *Suppose $Q = I$ and $A = A(\rho)$. Let*

$$\mathbf{d} = (I - \mathbf{u}\mathbf{u}')\mathbf{x}_0, \quad \mathbf{w}_\perp = (I - \mathbf{u}\mathbf{u}')\mathbf{w}.$$

¹³The vector $\mathbf{w}$ can represent persistent misinformation, agent-level bias, or anchored opinions.

Then every strict local optimum satisfies

$$\mathbf{b}^* \in \text{span}\{\mathbf{u}, \rho \mathbf{d}, \mathbf{w}_\perp\}.$$

Persistent bias therefore adds at most one relevant disagreement direction in this family. Average opinion lies along $\mathbf{u}$, inherited disagreement that survives one update lies along $\rho \mathbf{d}$, and the repeated source contributes $\mathbf{w}_\perp$. Every strict local optimum is therefore confined to this span.

The clique gives the sharpest case. When $\rho = 0$, inherited disagreement disappears after one update, but the disagreement supplied by $\mathbf{w}_\perp$ is replenished every period. Hence every strict local optimum satisfies

$$\mathbf{b}^* \in \text{span}\{\mathbf{u}, \mathbf{w}_\perp\}.$$

A persistent source can therefore create nonuniform targeting without creating high-dimensional disagreement, even when the network immediately eliminates inherited disagreement.

8 Conclusion

When an exposure rule must be reused, optimal targeting is not a ranking of influential agents. It is a matching problem between the opinion components the network preserves and the agents with leverage over them. Persistence determines what remains worth moving; leverage determines who can move it. In the U.S. Facebook climate network, this distinction shifts exposure toward Texas relative to authority-based targeting, bringing its optimal exposure close to California's.

The value of targeting flexibility is governed by the same endogenous path. Differences between coarse and full targeting matter only in directions where that path places weight. A single reusable rule can therefore perform nearly as well when the path is effectively low-dimensional, but loses value when recurrent shocks populate many new directions. Persistence alone does not make coarse targeting costly: a predictable bias can remain low-dimensional even when it never disappears.

The same logic applies to intermediate technologies – rules that adjust slowly, vary across groups, or emerge from influencer competition. The central design question is not how many

agents can be targeted, but whether the available exposure directions span the payoff-relevant disagreement the network carries forward.

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A Proofs

Some proofs for more standard or subsidiary results are relegated to Section S1 in the Supplementary appendix. These include Lemma 1, Lemma 2, Corollary 1, Lemma 3, Corollary 2 and Lemma 4.

A.1 Proofs from Section 3

Proof of Proposition 1: We derive the gradient of the value with respect to the exposure rule. Fix $\mathbf{b} \neq \mathbf{0}$. By Lemma 1, the optimal intervention level is unique and linear: there exists a vector $\mathbf{p}$ such that $r_t = -\mathbf{p}'\mathbf{x}_t$ for all $t \geq 0$. Under this policy, $\mathbf{x}_{t+1} = \tilde{A}(\mathbf{b})\mathbf{x}_t$ where $\tilde{A}(\mathbf{b}) \equiv A - \mathbf{b}\mathbf{p}'$. By Lemma 1, $\rho(\sqrt{\delta} \tilde{A}(\mathbf{b})) < 1$, so all discounted sums below are well defined. For any state $\mathbf{x}_t$, the Bellman equation evaluated at the optimal intervention level is

$$v(\mathbf{x}_t; \mathbf{b}) = -\mathbf{x}_t' Q \mathbf{x}_t - cr_t^2 - \delta \mathbf{x}_{t+1}' K^*(\mathbf{b}) \mathbf{x}_{t+1}, \text{ and } \mathbf{x}_{t+1} = A\mathbf{x}_t + \mathbf{b}r_t.$$

By the Envelope theorem, terms involving $\partial r_t / \partial \mathbf{b}$ vanish. Differentiating with respect to $\mathbf{b}$,

$$\nabla_{\mathbf{b}} v(\mathbf{x}_t; \mathbf{b}) = -2\delta r_t K^*(\mathbf{b}) \mathbf{x}_{t+1} + \delta \nabla_{\mathbf{b}} v(\mathbf{x}_{t+1}; \mathbf{b}).$$

Iterating forward from $t = 0$ and using $\rho(\sqrt{\delta} \tilde{A}(\mathbf{b})) < 1$ gives

$$\nabla_{\mathbf{b}} v(\mathbf{x}_0; \mathbf{b}) = -2\delta \sum_{t=0}^{\infty} \delta^t r_t K^*(\mathbf{b}) \mathbf{x}_{t+1}.$$

Substituting $r_t = -\mathbf{p}' \tilde{A}(\mathbf{b})^t \mathbf{x}_0$ and $\mathbf{x}_{t+1} = \tilde{A}(\mathbf{b})^{t+1} \mathbf{x}_0$ yields

$$\nabla_{\mathbf{b}} v(\mathbf{x}_0; \mathbf{b}) = 2\delta K^*(\mathbf{b}) \tilde{A}(\mathbf{b}) \left(\sum_{t=0}^{\infty} (\sqrt{\delta} \tilde{A}(\mathbf{b}))^t \mathbf{x}_0 \mathbf{x}_0' (\sqrt{\delta} \tilde{A}(\mathbf{b})')^t \right) \mathbf{p}.$$

Recall that $Z \equiv \sum_{t=0}^{\infty} (\sqrt{\delta} \tilde{A}(\mathbf{b}))^t \mathbf{x}_0 \mathbf{x}_0' (\sqrt{\delta} \tilde{A}(\mathbf{b})')^t$. Then $\nabla_{\mathbf{b}} v(\mathbf{x}_0; \mathbf{b}) = 2\delta K^*(\mathbf{b}) \tilde{A}(\mathbf{b}) Z \mathbf{p}$. Moreover, because $\rho(\sqrt{\delta} \tilde{A}(\mathbf{b})) < 1$, Z is the unique solution to the Lyapunov equation

$$Z = \delta \tilde{A}(\mathbf{b}) Z \tilde{A}(\mathbf{b})' + \mathbf{x}_0 \mathbf{x}_0'.$$

The Lagrangian for the constrained problem on the unit sphere is

$$\mathcal{L}(\mathbf{b}, \lambda) = v(\mathbf{x}_0; \mathbf{b}) - \lambda(\mathbf{b}'\mathbf{b} - 1).$$

At a constrained stationary point on the sphere, $\nabla_{\mathbf{b}} v(\mathbf{x}_0; \mathbf{b}) = 2\lambda \mathbf{b}$. Using the gradient expression above and absorbing the positive scalar 2δ into the multiplier gives $\mu \mathbf{b} = K^*(\mathbf{b}) \tilde{A}(\mathbf{b}) Z \mathbf{p}$, which is (7). At $\mathbf{b} = \mathbf{0}$, define

$$K^*(\mathbf{0}) = \sum_{t=0}^{\infty} \delta^t (A')^t Q A^t,$$

the no-intervention loss matrix. The fixed-rule value extends continuously to $\mathbf{b} = \mathbf{0}$. Hence $v(\mathbf{x}_0; \mathbf{b})$ is continuous on the compact feasible set $\{\mathbf{b} : \|\mathbf{b}\| \leq 1\}$. Existence follows from continuity of $v(\mathbf{x}_0; \mathbf{b})$ and compactness of the feasible set $\{\mathbf{b} : \|\mathbf{b}\| \leq 1\}$. If $Q A \mathbf{x}_0 \neq \mathbf{0}$, Lemma 2 implies that every optimizer satisfies $\|\mathbf{b}^*\| = 1$. Hence every optimizer satisfies the first-order condition above. $\square$

A.2 Proofs from Section 4

Proof of Proposition 2: Let $\Pi = I - \mathbf{u}\mathbf{u}'$ denote the projection onto deviations from average opinion. Decompose any exposure rule as $\mathbf{b} = \alpha \mathbf{u} + \mathbf{q}$, $\alpha = \mathbf{u}'\mathbf{b}$, and $\mathbf{q} \perp \mathbf{u}$. Recall $\bar{x}_t = \frac{1}{n} \mathbf{1}' \mathbf{x}_t = \frac{1}{\sqrt{n}} \mathbf{u}' \mathbf{x}_t$. Because A is doubly stochastic, $A\mathbf{u} = \mathbf{u}$ and $\mathbf{u}'A = \mathbf{u}'$. Therefore, under $(\mathbf{b}, \{r_t\})$, average opinion evolves according to

$$\bar{x}_{t+1} = \bar{x}_t + \frac{\alpha}{\sqrt{n}} r_t.$$

Suppose first that $Ad = \mathbf{0}$. Given any feasible exposure rule $\mathbf{b} = \alpha \mathbf{u} + \mathbf{q}$ and intervention sequence $\{r_t\}$, consider uniform exposure $\mathbf{u}$ with $\hat{r}_t = \alpha r_t$. Thus $\bar{x}_{t+1} = \bar{x}_t + \frac{1}{\sqrt{n}} \hat{r}_t$. The two policies generate the same path of average opinion. Moreover, $Ad = \mathbf{0}$, so inherited disagreement disappears after the first update. Uniform exposure creates no new

disagreement, whereas a nonuniform exposure rule may do so. Since

$$\|\mathbf{x}_t\|^2 = n\bar{x}_t^2 + \|\Pi\mathbf{x}_t\|^2,$$

uniform exposure therefore generates weakly lower opinion loss at every date. Its intervention cost is also weakly lower because $\hat{r}_t^2 = \alpha^2 r_t^2 \leq r_t^2$, where $|\alpha| \leq \|\mathbf{b}\| \leq 1$. Thus uniform targeting weakly dominates every exposure rule and is optimal.

Conversely, suppose that $A\mathbf{d} \neq \mathbf{0}$ and consider uniform exposure, $\mathbf{b} = \mathbf{u}$. Since both $\text{span}\{\mathbf{u}\}$ and $\mathbf{u}^\perp$ are invariant under A and A' , the Riccati map preserves matrices that do not mix the average-opinion direction and the disagreement space. By uniqueness of the stabilizing Riccati solution,

$$K^* = \tilde{k}_1 \mathbf{u}\mathbf{u}' + K_\perp, \quad K_\perp = \Pi K^* \Pi,$$

where $K_\perp$ is positive definite on $\mathbf{u}^\perp$. Moreover,

$$\mathbf{p} = p_1 \mathbf{u}, \quad \tilde{A}(\mathbf{u}) = a_1 \mathbf{u}\mathbf{u}' + A_\perp, \quad A_\perp = \Pi A \Pi,$$

where

$$p_1 = \frac{\delta \tilde{k}_1}{c + \delta \tilde{k}_1} > 0, \quad a_1 = 1 - p_1 = \frac{c}{c + \delta \tilde{k}_1} \in (0, 1).$$

Recall $\bar{x}_0 = \frac{1}{n} \mathbf{1}' \mathbf{x}_0$, $\mathbf{d} = \Pi \mathbf{x}_0$ so that $\mathbf{u}' \mathbf{x}_0 = \sqrt{n} \bar{x}_0$. Since

$$\tilde{A}(\mathbf{u}) = a_1 \mathbf{u}\mathbf{u}' + A_\perp,$$

we have

$$\Pi \tilde{A}(\mathbf{u})^t \mathbf{x}_0 = A_\perp^t \mathbf{d}, \quad \mathbf{x}_0' (\tilde{A}(\mathbf{u})')^t \mathbf{u} = a_1^t \sqrt{n} \bar{x}_0.$$

Hence

$$\Pi Z \mathbf{u} = \sqrt{n} \bar{x}_0 (I - \delta a_1 A_\perp)^{-1} \mathbf{d}.$$

Because A is doubly stochastic, $\|A\|_2 \leq \sqrt{\|A\|_1 \|A\|_\infty} = 1$. Hence $\rho(A_\perp) \leq \|A_\perp\|_2 \leq 1$. Since $\delta a_1 < 1$, the matrix $I - \delta a_1 A_\perp$ is invertible. If uniform exposure were optimal, Proposition 1 would imply that it satisfies (7). Projecting the right-hand side of (7) onto $\mathbf{u}^\perp$ gives

$$\Pi K^* \tilde{A}(\mathbf{u}) Z \mathbf{p} = \sqrt{n} \bar{x}_0 p_1 K_\perp A_\perp (I - \delta a_1 A_\perp)^{-1} \mathbf{d}.$$

Since $K_\perp$ and $I - \delta a_1 A_\perp$ are invertible on $\mathbf{u}^\perp$, this expression is zero if and only if $A_\perp \mathbf{d} = A \mathbf{d} = \mathbf{0}$. Thus when $A \mathbf{d} \neq \mathbf{0}$, its right-hand side has a nonzero disagreement component, whereas $\mathbf{b} = \mathbf{u}$ has none. Uniform exposure therefore cannot be optimal. $\square$

Proof of Proposition 3: By Lemma 2, an optimal exposure rule satisfies $\|\mathbf{b}\| = 1$. Fix such a rule $\mathbf{b}$. Since $A = \mathbf{1}\mathbf{a}'$, the Riccati equation implies that

$$K = I + \nu \mathbf{a}\mathbf{a}'$$

for some scalar $\nu > 0$. Hence $\mathbf{x}_0' K \mathbf{x}_0 = \|\mathbf{x}_0\|^2 + \nu (\mathbf{a}' \mathbf{x}_0)^2$. For every $\mathbf{x}_0$ with $\mathbf{a}' \mathbf{x}_0 \neq 0$, the optimal exposure rule therefore minimizes the same scalar ν and is independent of $\mathbf{x}_0$. Substituting $K = I + \nu \mathbf{a}\mathbf{a}'$ into the Riccati equation gives

$$\nu = \delta \left[n + \nu - \delta \frac{((\mathbf{1} + \nu \mathbf{a})' \mathbf{b})^2}{\mathbf{b}'((c + \delta)I + \delta \nu \mathbf{a}\mathbf{a}')\mathbf{b}} \right].$$

At an optimum, $\nu(\mathbf{b})$ is minimized over the unit sphere. Hence, for every tangent direction $\mathbf{h}$ satisfying $(\mathbf{b}^*)' \mathbf{h} = 0$, the directional derivative of $\nu(\mathbf{b})$ at $\mathbf{b}^*$ vanishes. Write the scalar Riccati equation as

$$F(\mathbf{b}, \nu) \equiv (1 - \delta)\nu - \delta n + \delta^2 R(\mathbf{b}, \nu) = 0, \text{ where } R(\mathbf{b}, \nu) = \frac{((\mathbf{1} + \nu \mathbf{a})' \mathbf{b})^2}{\mathbf{b}'((c + \delta)I + \delta \nu \mathbf{a}\mathbf{a}')\mathbf{b}}.$$

Differentiating $F(\mathbf{b}, \nu(\mathbf{b})) = 0$ in a tangent direction at $\mathbf{b}^*$ and using the vanishing derivative of $\nu(\mathbf{b})$ gives

$$D_{\mathbf{b}} R(\mathbf{b}^*, \nu^*)[\mathbf{h}] = 0.$$

Thus, for its associated value ν^* , $\mathbf{b}^*$ is a constrained critical point of $R(\mathbf{b}, \nu^*)$. For any positive definite matrix M and nonzero vector $\mathbf{y}$, the constrained critical points of $\frac{(\mathbf{y}' \mathbf{b})^2}{\mathbf{b}' M \mathbf{b}}$ either satisfy $\mathbf{y}' \mathbf{b} = 0$ or are proportional to $M^{-1} \mathbf{y}$. The former are global minima and the latter are global maxima. In the present problem, $M = (c + \delta)I + \delta \nu \mathbf{a}\mathbf{a}'$ and $\mathbf{y} = \mathbf{1} + \nu \mathbf{a}$. The zero-numerator case implies $A' K \mathbf{b} = \mathbf{0}$, so the optimal intervention is identically zero. Since $\mathbf{a}' \mathbf{x}_0 \neq 0$, Lemma 2 rules out this case. Hence

$$\mathbf{b}^* \propto ((c + \delta)I + \delta \nu \mathbf{a}\mathbf{a}')^{-1} (\mathbf{1} + \nu \mathbf{a}).$$

By the Sherman–Morrison formula,

$$\mathbf{b}^* \propto \mathbf{1} + \frac{\nu c}{c + \delta + \delta \nu \|\mathbf{a}\|^2} \mathbf{a}.$$

The coefficient on $\mathbf{a}$ is strictly positive. Since $\mathbf{b}$ and $-\mathbf{b}$ are equivalent exposure rules, choose the normalization for which the coefficient on $\mathbf{1}$ is positive. Then

$$\mathbf{b}^* = \eta \mathbf{1} + \kappa \mathbf{a}, \quad \eta > 0, \quad \kappa > 0.$$

Consequently, $b_i^* - b_j^* = \kappa(a_i - a_j)$. Since $a_1 > \dots > a_n > 0$, it follows that $b_1^* > \dots > b_n^* > 0$. $\square$

Proof of Proposition 4: Work in the eigenbasis of A_0 and let $P = (\mathbf{p}_1, \dots, \mathbf{p}_n)$ where $\mathbf{p}_1 = \mathbf{u}$. Thus $\mathbf{e}_j$ denotes the j th coordinate vector in the transformed basis. Let

$$\tilde{\tilde{A}} = P' \tilde{A} P$$

denote the induced opinion-transition matrix in the eigenbasis. Then $\tilde{\tilde{A}}_0 = \text{diag}(\tau, \lambda_2, \dots, \lambda_n)$. By scale invariance, any initial state near the uniform-opinion benchmark can be normalized as

$$\tilde{\mathbf{x}}_0 = \mathbf{e}_1 + \boldsymbol{\xi}, \quad \boldsymbol{\xi} \perp \mathbf{e}_1.$$

Parameterize unit exposure rules near $\mathbf{e}_1$ by

$$\tilde{\mathbf{b}}(\boldsymbol{\zeta}) = \sqrt{1 - \|\boldsymbol{\zeta}\|^2} \mathbf{e}_1 + \boldsymbol{\zeta}, \quad \boldsymbol{\zeta} \perp \mathbf{e}_1.$$

Hence

$$\tilde{\mathbf{b}}(\boldsymbol{\zeta}) = \mathbf{e}_1 + \boldsymbol{\zeta} + O(\|\boldsymbol{\zeta}\|^2).$$

At the benchmark $(\boldsymbol{\zeta}, \boldsymbol{\xi}, \varepsilon) = (\mathbf{0}, \mathbf{0}, 0)$, the Riccati matrix $\tilde{K}_0 = \text{diag}(\tilde{k}_1, \dots, \tilde{k}_n)$. The closed-loop matrix is $\tilde{\tilde{A}}_0 = \text{diag}(\tau, \lambda_2, \dots, \lambda_n)$. Moreover, $\tilde{\mathbf{p}}_0 = \ell_1 \mathbf{e}_1$ where

$$\ell_1 = \frac{\delta \tilde{k}_1}{c + \delta \tilde{k}_1}, \quad \tau = 1 - \ell_1 = \frac{c}{c + \delta \tilde{k}_1} \in (0, 1),$$

and, for $j > 1$,

$$\tilde{k}_j = \frac{1}{1 - \delta\lambda_j^2}.$$

Also,

$$\tilde{Z}_0 = z_1 \mathbf{e}_1 \mathbf{e}_1', \quad z_1 = \frac{1}{1 - \delta\tau^2}.$$

We first compute the tangent Hessian common to both perturbations. For each $j > 1$, consider the one-dimensional exposure rotation

$$\tilde{\mathbf{b}}_j(\gamma) = \sqrt{1 - \gamma^2} \mathbf{e}_1 + \gamma \mathbf{e}_j$$

and write

$$\tilde{K}_{1j}(\gamma) = q_j \gamma + o(\gamma), \quad \tilde{K}_{11}(\gamma) = \tilde{k}_1 + h_j \gamma^2 + o(\gamma^2).$$

Expanding the $(1, j)$ and $(1, 1)$ entries of the transformed Riccati equation gives

$$q_j = -\frac{\delta\lambda_j \ell_1 \tilde{k}_j}{1 - \delta\tau\lambda_j}$$

and

$$h_j = \frac{\delta \left[\ell_1 \tau (\tilde{k}_1 - 2q_j) + \ell_1^2 \tilde{k}_j \right]}{1 - \delta\tau^2}.$$

Because $|\lambda_j| \leq 1$,

$$1 - \delta\tau\lambda_j > 0, \quad 1 + \delta\tau\lambda_j > 0,$$

and

$$\tau(\tilde{k}_1 - 2q_j) + \ell_1 \tilde{k}_j = \frac{\tau \tilde{k}_1 (1 - \delta\tau\lambda_j) + \ell_1 \tilde{k}_j (1 + \delta\tau\lambda_j)}{1 - \delta\tau\lambda_j} > 0.$$

Hence $h_j > 0$ for every $j > 1$.

At the benchmark, reversing the sign of any disagreement coordinate leaves D , $\mathbf{e}_1$, and $Q = I$ unchanged. Hence value is invariant under $\zeta_j \mapsto -\zeta_j$ for every $j > 1$. It is therefore even in each ζ_j , which rules out all quadratic cross terms $\zeta_j \zeta_l$, $j \neq l$. The constrained Hessian of value with respect to ζ is consequently $-2 \text{diag}(h_2, \dots, h_n)$. Because $h_j > 0$ for every $j > 1$, this Hessian is negative definite, so $\zeta = \mathbf{0}$ is a nondegenerate strict local

maximum at the benchmark. Let

$$\Psi(\zeta; \xi, \varepsilon) = D_\zeta \tilde{v}(\mathbf{e}_1 + \xi; \tilde{\mathbf{b}}(\zeta), D + \varepsilon \tilde{B})$$

denote the first-order condition in the local coordinates. This map is continuously differentiable, satisfies $\Psi(\mathbf{0}; \mathbf{0}, 0) = \mathbf{0}$, and has

$$D_\zeta \Psi(\mathbf{0}; \mathbf{0}, 0) = -2 \operatorname{diag}(h_2, \dots, h_n),$$

which is nonsingular. The implicit-function theorem therefore yields a unique continuously differentiable branch $\zeta^*(\xi, \varepsilon)$ near $(\mathbf{0}, 0)$, with $\zeta^*(\mathbf{0}, 0) = \mathbf{0}$. The next two calculations give its partial derivatives at the benchmark; the joint expansion follows from its first-order Taylor approximation.

We now differentiate the local optimum with respect to initial disagreement, holding $\varepsilon = 0$. The terms in the local loss that depend on ζ are

$$\sum_{j>1} (h_j \zeta_j^2 + 2q_j \zeta_j \xi_j) + o(\|\zeta\|^2 + \|\zeta\| \|\xi\|).$$

The first-order condition in direction j is therefore

$$h_j \zeta_j + q_j \xi_j + o(\|\zeta\| + \|\xi\|) = 0.$$

Since $h_j > 0$,

$$\frac{\partial \zeta_j^*}{\partial \xi_j}(\mathbf{0}, 0) = -\frac{q_j}{h_j} \equiv M(\lambda_j),$$

while

$$\frac{\partial \zeta_j^*}{\partial \xi_l}(\mathbf{0}, 0) = 0 \quad \text{for } l \neq j.$$

Let

$$\tilde{k}(\lambda) = \frac{1}{1 - \delta \lambda^2}.$$

Substituting the expressions for q_j and h_j gives

$$M(\lambda) = \frac{\lambda \tilde{k}(\lambda)(1 - \delta \tau^2)}{\tau \tilde{k}_1(1 - \delta \tau \lambda) + \ell_1 \tilde{k}(\lambda)(1 + \delta \tau \lambda)}.$$

The denominator is strictly positive for every $\lambda \in [-1, 1]$. Hence $M(0) = 0$ and $M(\lambda)$ has the sign of λ . Direct differentiation shows that M is strictly increasing on $[-1, 1]$.

We next differentiate with respect to the network perturbation, holding $\xi = 0$. Let

$$\tilde{B} = P'BP, \quad \tilde{s} = \tilde{B}'e_1 = P's, \quad s = B'u.$$

Because $B\mathbf{1} = 0$,

$$\tilde{B}e_1 = 0 \quad \text{and} \quad e_1'\tilde{s} = 0.$$

Holding $\tilde{b} = e_1$, define

$$\dot{\tilde{K}} = \left. \frac{\partial \tilde{K}^*(e_1; D + \varepsilon \tilde{B})}{\partial \varepsilon} \right|_{\varepsilon=0}$$

and let

$$\tilde{k} = (I - e_1 e_1') \dot{\tilde{K}} e_1.$$

Differentiating the transformed Riccati equation at the benchmark and projecting its first column onto $e_1^\perp$ gives

$$\tilde{k} = \delta\tau \tilde{k}_1 \tilde{B}'e_1 + \delta\tau D\tilde{k}.$$

Hence

$$\tilde{k} = \delta\tau \tilde{k}_1 (I - \delta\tau D)^{-1} \tilde{B}'e_1.$$

Define

$$g = (I - \delta\tau A_0')^{-1}s = \sum_{j>1} g_j p_j.$$

Because $e_1'\tilde{s} = 0$ and D preserves $e_1^\perp$, we also have $g_1 = 0$. Since

$$g_j = \frac{\tilde{s}_j}{1 - \delta\tau \lambda_j}, \quad \tilde{s}_j = e_j' \tilde{B}'e_1,$$

we obtain, for every $j > 1$,

$$\dot{\tilde{K}}_{j1} = \delta\tau \tilde{k}_1 g_j.$$

We now differentiate the targeting condition. In the eigenbasis, the value gradient with respect to the exposure rule is

$$\nabla_{\tilde{b}} \tilde{v} = 2\delta \tilde{K} \tilde{A} \tilde{Z} \tilde{p}.$$

At the benchmark, this vector lies in $\text{span}\{\mathbf{e}_1\}$. Holding $\tilde{\mathbf{b}} = \mathbf{e}_1$, differentiation with respect to ε gives, for each $j > 1$,

$$\left. \frac{\partial}{\partial \varepsilon} \mathbf{e}'_j \nabla_{\tilde{\mathbf{b}}} \tilde{v} \right|_{(\boldsymbol{\xi}, \varepsilon) = (\mathbf{0}, 0)} = 2\delta\tau z_1 \ell_1 \dot{K}_{j1}.$$

Since $\tilde{B}\mathbf{e}_1 = \mathbf{0}$, the remaining benchmark factors map $\mathbf{e}_1$ into $\text{span}\{\mathbf{e}_1\}$. Hence the terms involving the derivatives of $\tilde{A}$, $\tilde{Z}$, and $\tilde{\mathbf{p}}$ remain in $\text{span}\{\mathbf{e}_1\}$ and vanish after projection onto $\mathbf{e}_j$.

The change in the exposure rule contributes through the tangent Hessian. Differentiating the projected first-order condition along the local optimum therefore gives

$$-2h_j \frac{\partial \zeta_j^*}{\partial \varepsilon}(\mathbf{0}, 0) + 2\delta\tau z_1 \ell_1 \dot{K}_{j1} = 0.$$

Using

$$\dot{K}_{j1} = \delta\tau \tilde{k}_1 g_j \quad \text{and} \quad z_1 = \frac{1}{1 - \delta\tau^2},$$

we obtain

$$\frac{\partial \zeta_j^*}{\partial \varepsilon}(\mathbf{0}, 0) = \chi_j g_j,$$

where

$$\chi_j = \frac{\delta^2 \tau^2 \ell_1 \tilde{k}_1}{(1 - \delta\tau^2) h_j} > 0.$$

Taking the total differential of the jointly differentiable branch gives, for each $j > 1$,

$$\zeta_j^*(\boldsymbol{\xi}, \varepsilon) = M(\lambda_j) \xi_j + \varepsilon \chi_j g_j + o(\|\boldsymbol{\xi}\| + |\varepsilon|).$$

Since

$$\tilde{\mathbf{b}}(\boldsymbol{\zeta}) = \mathbf{e}_1 + \boldsymbol{\zeta} + O(\|\boldsymbol{\zeta}\|^2),$$

it follows that

$$\tilde{\mathbf{b}}^*(\boldsymbol{\xi}, \varepsilon) = \mathbf{e}_1 + \sum_{j>1} [M(\lambda_j) \xi_j + \varepsilon \chi_j g_j] \mathbf{e}_j + o(\|\boldsymbol{\xi}\| + |\varepsilon|).$$

Thus persistence and leverage add to first order; their interaction enters only at higher order.

Finally, we verify the clique claim following Proposition 4. Let $A_0 = J/n$ and consider the perturbation

$$B = \mathbf{1} \left(\mathbf{a} - \frac{1}{n} \mathbf{1} \right)'.$$

Then

$$\mathbf{s} = B' \mathbf{u} = \sqrt{n} \left(\mathbf{a} - \frac{1}{n} \mathbf{1} \right).$$

Because $\mathbf{s} \perp \mathbf{1}$ and $A_0' \mathbf{s} = \mathbf{0}$,

$$\mathbf{g} = (I - \delta\tau A_0')^{-1} \mathbf{s} = \mathbf{s}.$$

Moreover, permutation symmetry implies that $\chi_j = \chi > 0$ for every $j > 1$. Setting $\boldsymbol{\xi} = \mathbf{0}$ and returning to the original coordinates therefore gives

$$\mathbf{b}^*(\varepsilon) = \mathbf{u} + \varepsilon \chi \sqrt{n} \left(\mathbf{a} - \frac{1}{n} \mathbf{1} \right) + o(|\varepsilon|).$$

Equivalently, for some $C > 0$,

$$\mathbf{b}^*(\varepsilon) = \mathbf{u} + \varepsilon C \left(\mathbf{a} - \frac{1}{n} \mathbf{1} \right) + o(|\varepsilon|).$$

Hence $b_i^*(\varepsilon) - b_j^*(\varepsilon) = \varepsilon C(a_i - a_j) + o(|\varepsilon|)$, which recovers locally the pairwise authority pattern in Proposition 3. $\square$

A.3 Proofs from Section 5

Proof of Proposition 5: first, we characterize the payoff gap in a simple lemma. The proof is in Supplementary Appendix S1.

Lemma 4. *Let $\mathbf{b} \in S^{n-1}$ be any coarse exposure rule. The optimality gap from coarse targeting under $\mathbf{b}$ to full targeting can be expressed as*

$$G(\mathbf{b}) = \sum_{t=0}^{\infty} \delta^t \tilde{A}'^t \Phi(\mathbf{b}) \tilde{A}^t, \quad \Delta(\mathbf{b}) = \mathbf{x}_0' G(\mathbf{b}) \mathbf{x}_0, \quad (9)$$

where $\Phi(\mathbf{b}) \equiv (\mathbf{b}\mathbf{p}' - F^f)'(cI + \delta K^f)(\mathbf{b}\mathbf{p}' - F^f) \geq 0$ is the gap source matrix.

By Lemma 4, $G(\mathbf{b}) = \sum_{t=0}^{\infty} \delta^t \tilde{A}^{t'} \Phi(\mathbf{b}) \tilde{A}^t$ and $\Delta(\mathbf{b}) = \mathbf{x}_0' G(\mathbf{b}) \mathbf{x}_0$. Therefore,

$$\Delta(\mathbf{b}) = \sum_{t=0}^{\infty} \delta^t \mathbf{x}_0' \tilde{A}^{t'} \Phi(\mathbf{b}) \tilde{A}^t \mathbf{x}_0 = \sum_{t=0}^{\infty} \delta^t \text{tr} \left(\Phi(\mathbf{b}) \tilde{A}^t \mathbf{x}_0 \mathbf{x}_0' \tilde{A}^{t'} \right) = \text{tr} \left(\Phi(\mathbf{b}) \sum_{t=0}^{\infty} \delta^t \tilde{A}^t \mathbf{x}_0 \mathbf{x}_0' \tilde{A}^{t'} \right).$$

By definition, $Z(\mathbf{b}) = \sum_{t=0}^{\infty} \delta^t \tilde{A}^t \mathbf{x}_0 \mathbf{x}_0' \tilde{A}^{t'}$. Thus $\Delta(\mathbf{b}) = \text{tr}(\Phi(\mathbf{b}) Z(\mathbf{b}))$. $\square$

A.4 Proofs from Section 7

Proof of Proposition 6: We first compare myopic and full targeting. Matrix $\tilde{K}^f$ is the unique diagonal pd matrix solution to

$$\tilde{K}^f = I + \delta c D (\delta \tilde{K}^f + c I)^{-1} \tilde{K}^f D. \quad (10)$$

Matrix $\tilde{K}^f$ captures the decision-maker's future discounted payoff, net of the cost of the optimal policy. Let $\tilde{K}_{jj}^f$ be the j th diagonal entry, which strictly increases in $|\lambda_j|$. Let $\Sigma = \sigma^2 I$. Since the eigenbasis transformation is orthogonal, the shock covariance remains Σ in the transformed coordinates. The decision-maker's value consists of two parts:

$$\tilde{v}^f(\tilde{\mathbf{x}}_0) = -\tilde{\mathbf{x}}_0' \tilde{K}^f \tilde{\mathbf{x}}_0 - \sum_{t=0}^{\infty} \delta^{t+1} \mathbb{E} \left(\tilde{\nu}_t' \tilde{K}^f \tilde{\nu}_t \right) = -\tilde{\mathbf{x}}_0' \tilde{K}^f \tilde{\mathbf{x}}_0 - \frac{\delta}{1-\delta} \text{tr}(\tilde{K}^f \Sigma). \quad (11)$$

For myopic targeting, without noise, the value is

$$-\tilde{\mathbf{x}}_0' \left(I - \frac{\delta c^2}{(\delta + c)^2} D^2 \right)^{-1} \left(I + \frac{c \delta^2}{(\delta + c)^2} D^2 \right) \tilde{\mathbf{x}}_0 = -\tilde{\mathbf{x}}_0' \tilde{K}^m \tilde{\mathbf{x}}_0, \quad (12)$$

where $\tilde{K}^m$ is the product of two matrices above, the first matrix captures how each $\tilde{\mathbf{x}}_t$ depends on $\tilde{\mathbf{x}}_0$ given the optimal policy, and the second matrix is how each stage payoff depends on $\tilde{\mathbf{x}}_t$. Clearly, $\tilde{K}^m$ is diagonal with entry $\tilde{K}_{jj}^m$, which strictly increases in $|\lambda_j|$. Each eigen-dimension is an independent scalar linear-quadratic problem. Full targeting uses the unique dynamically optimal scalar feedback, whereas the myopic feedback differs from it whenever $\lambda_j \neq 0$. Strict convexity of the scalar Bellman objective therefore implies $\tilde{k}_{jj}^f < \tilde{k}_{jj}^m$ whenever $\lambda_j \neq 0$, with equality when $\lambda_j = 0$. Moreover, the scalar Riccati equation implies that $\tilde{k}_{jj}^f$ is increasing in λ_j^2 . Since $|\lambda_j| \leq 1 = \lambda_1$, we have $\tilde{k}_{jj}^f \leq \tilde{k}_{11}^f$. With

noise, we have

$$\tilde{v}^m(\tilde{\mathbf{x}}_0) = -\tilde{\mathbf{x}}_0' \tilde{K}^m \tilde{\mathbf{x}}_0 - \frac{\delta}{1-\delta} \text{tr}(\tilde{K}^m \Sigma).$$

Let

$$\alpha = \frac{\delta \sigma^2}{1-\delta}, \quad \text{and} \quad \eta_j = \frac{\tilde{k}_{jj}^m - \tilde{k}_{jj}^f}{\tilde{k}_{jj}^f} \geq 0$$

for each j . The inequality is strict whenever $\lambda_j \neq 0$. Since

$$|\tilde{v}^f| = \sum_{j=1}^n ((\tilde{x}_0^j)^2 + \alpha) \tilde{k}_{jj}^f \quad \text{and} \quad |\tilde{v}^m| = \sum_{j=1}^n ((\tilde{x}_0^j)^2 + \alpha) \tilde{k}_{jj}^f (1 + \eta_j),$$

we can write

$$\frac{\tilde{v}^f}{\tilde{v}^m} = \frac{1}{1 + \sum_{j=1}^n \pi_j \eta_j},$$

where

$$\pi_j = \frac{((\tilde{x}_0^j)^2 + \alpha) \tilde{k}_{jj}^f}{\sum_{l=1}^n ((\tilde{x}_0^l)^2 + \alpha) \tilde{k}_{ll}^f}, \quad \sum_{j=1}^n \pi_j = 1.$$

Let $\bar{\eta} = \max_j \eta_j$. Because A is row stochastic, $\lambda_1 = 1$, so $\eta_1 > 0$. Moreover, $\tilde{k}_{jj}^f \leq \tilde{k}_{11}^f$ for every j , and therefore

$$\pi_1 \geq \frac{\alpha}{\|\tilde{\mathbf{x}}_0\|^2 + n\alpha}.$$

Define $\underline{\eta} = \frac{\alpha}{\|\tilde{\mathbf{x}}_0\|^2 + n\alpha} \eta_1 > 0$. It follows that

$$\underline{\eta} \leq \sum_{j=1}^n \pi_j \eta_j \leq \bar{\eta}, \quad \text{or} \quad \frac{1}{1 + \bar{\eta}} \leq \left| \frac{\tilde{v}^f}{\tilde{v}^m} \right| \leq \frac{1}{1 + \underline{\eta}}.$$

In particular, full targeting strictly dominates myopic targeting.

Second, we compare full and coarse targeting. Suppose first that $\text{rank}(A) = 1$. Since A is symmetric and row stochastic, $A = J/n$. Hence, in the eigenbasis of A , $D = \text{diag}(1, 0, \dots, 0)$. Under full targeting, the optimal feedback matrix is diagonal and has zero entries in every disagreement dimension. Thus full targeting intervenes only in the average-opinion dimension. The uniform coarse exposure rule $\tilde{\mathbf{b}} = \mathbf{e}_1$ implements exactly the same intervention. In the first dimension, the coarse and full Riccati equations coincide, while for every $j > 1$, $\tilde{k}_{jj}^c = \tilde{k}_{jj}^f = 1$. Therefore $\tilde{K}^c = \tilde{K}^f$ and $\tilde{v}^c(\tilde{\mathbf{x}}_0) = \tilde{v}^f(\tilde{\mathbf{x}}_0)$. Now suppose $\text{rank}(A) > 1$. Full targeting weakly dominates coarse targeting because every

coarse intervention $\tilde{\mathbf{b}}r_t$ is feasible under full targeting at the same cost. Under full targeting,

$$\tilde{\mathbf{u}}_t^f = -\tilde{F}^f \tilde{\mathbf{x}}_t, \quad \tilde{F}^f = \delta(cI + \delta\tilde{K}^f)^{-1}\tilde{K}^f D.$$

Since $\tilde{K}^f$ is pd and diagonal, $\text{rank}(\tilde{F}^f) = \text{rank}(D) = \text{rank}(A) > 1$. Thus the full-targeting intervention does not always lie in any fixed one-dimensional subspace. By contrast, under any coarse exposure rule $\tilde{\mathbf{b}}$, every intervention lies in $\text{span}\{\tilde{\mathbf{b}}\}$. For a fixed $\tilde{\mathbf{b}}$, coarse targeting can therefore match the full-targeting intervention only at states satisfying $\tilde{F}^f \tilde{\mathbf{x}}_t \in \text{span}\{\tilde{\mathbf{b}}\}$. Because $\text{rank}(\tilde{F}^f) > 1$, this set is a proper subspace of $\mathbb{R}^n$. Since $\sigma > 0$, the state has full support, so it lies outside this subspace with positive probability. At those states, no scalar r_t can reproduce the full-targeting intervention. Since the Bellman objective is strictly concave in the intervention vector, coarse targeting then yields a strictly lower payoff. Thus full targeting strictly dominates coarse targeting whenever $\text{rank}(A) > 1$. $\square$

Proof of Proposition 7: We show each of the two claims in turn.

Step 1: An optimal coarse exposure rule can be chosen in $\text{span}\{\mathbf{u}, \mathbf{d}\}$. Choose the eigenbasis so that its second vector is aligned with $\mathbf{d}$. Let

$$d = \|\mathbf{d}\|, \quad P = \left(\mathbf{u}, \frac{\mathbf{d}}{d}, \mathbf{p}_3, \dots, \mathbf{p}_n \right).$$

Then $\tilde{\mathbf{x}}_0 = P' \mathbf{x}_0 = (\sqrt{n} \bar{x}_0, d, 0, \dots, 0)'$ and $D = \text{diag}(1, \rho, \dots, \rho)$. Take any optimal exposure rule and write $\tilde{\mathbf{b}} = a\mathbf{e}_1 + b\mathbf{g}$, $\mathbf{g} \perp \mathbf{e}_1$ and $\|\mathbf{g}\| = 1$, where $b \geq 0$. If $b = 0$, the claim is true. Fixing a, b , the cost matrix has the block form

$$\tilde{K}^c(\mathbf{g}) = \begin{pmatrix} \bar{K}^c(a, b) & 0 \\ 0 & \tilde{k}_\rho^\emptyset I_{n-2} \end{pmatrix}$$

relative to the decomposition into $\text{span}\{\mathbf{e}_1, \mathbf{g}\}$ and its orthogonal complement. Because all disagreement directions have the same eigenvalue and shocks are isotropic, $\bar{K}^c(a, b)$ and $\text{tr} \tilde{K}^c(\mathbf{g})$ do not depend on the particular choice of $\mathbf{g}$.

Let $s = de_2' \mathbf{g}$. The initial disagreement orthogonal to $\mathbf{g}$ has squared norm $d^2 - s^2$, so, up to terms independent of $\mathbf{g}$,

$$\tilde{v}(s) = -2\bar{K}_{12}^c(a, b)\sqrt{n} \bar{x}_0 s - \left(\bar{K}_{22}^c(a, b) - \tilde{k}_\rho^\emptyset \right) s^2.$$

Because zero intervention is feasible,

$$\bar{K}_{22}^c(a, b) \leq \tilde{k}_\rho^\emptyset.$$

Hence $\tilde{v}^c(s)$ is convex on $[-d, d]$ and is maximized at $s = d$ or $s = -d$. Thus an optimal disagreement direction can be chosen as $\mathbf{g} = \mathbf{e}_2$ or $\mathbf{g} = -\mathbf{e}_2$. Therefore an optimal exposure rule can be chosen in $\text{span}\{\mathbf{e}_1, \mathbf{e}_2\}$, or equivalently in $\text{span}\{\mathbf{u}, \mathbf{d}\}$.

Step 2: Asymptotic ranking. Consider a sequence of economies as $n \rightarrow \infty$. In the basis constructed in Step 1, $\tilde{\mathbf{x}}_0^{(n)} = (\mathbf{u}'_n \mathbf{x}_0^{(n)}, \|\mathbf{d}_n\|, 0, \dots, 0)'$. By assumption, $\|\tilde{\mathbf{x}}_0^{(n)}\| = O(1)$. For every disagreement dimension $j > 1$, the scalar cost coefficients under full targeting, myopic targeting, and no intervention are respectively $\tilde{k}_\rho^f$, $\tilde{k}_\rho^m$, and $\tilde{k}_\rho^\emptyset$. Then

$$\tilde{k}_\rho^f = 1 + \delta c \rho^2 \frac{\tilde{k}_\rho^f}{c + \delta \tilde{k}_\rho^f}, \quad \tilde{k}_\rho^m = \frac{1 + \frac{c \delta^2 \rho^2}{(c + \delta)^2}}{1 - \frac{\delta c^2 \rho^2}{(c + \delta)^2}}, \quad \tilde{k}_\rho^\emptyset = \frac{1}{1 - \delta \rho^2}.$$

Since $\rho \neq 0$,

$$0 < \tilde{k}_\rho^f < \tilde{k}_\rho^m < \tilde{k}_\rho^\emptyset,$$

because full targeting strictly improves on myopic targeting, and myopic targeting strictly improves on no intervention. Under full targeting,

$$-\tilde{v}_n^f = \tilde{k}_1^f (\mathbf{u}'_n \mathbf{x}_0^{(n)})^2 + \tilde{k}_\rho^f \|\mathbf{d}_n\|^2 + \frac{\delta \sigma^2}{1 - \delta} \left(\tilde{k}_1^f + (n - 1) \tilde{k}_\rho^f \right).$$

Similarly, under myopic targeting,

$$-\tilde{v}_n^m = \tilde{k}_1^m (\mathbf{u}'_n \mathbf{x}_0^{(n)})^2 + \tilde{k}_\rho^m \|\mathbf{d}_n\|^2 + \frac{\delta \sigma^2}{1 - \delta} \left(\tilde{k}_1^m + (n - 1) \tilde{k}_\rho^m \right).$$

The initial-state terms remain bounded by assumption. For coarse targeting, since the optimal exposure rule has support only on the first two coordinates,

$$\tilde{K}_n^c = \begin{pmatrix} \bar{K}_n^c & 0 \\ 0 & \tilde{k}_\rho^\emptyset I_{n-2} \end{pmatrix},$$

where $\bar{K}_n^c$ is the leading 2×2 block. Then

$$-\tilde{v}_n^c = \left(\frac{\mathbf{u}'_n \mathbf{x}_0^{(n)}}{\|\mathbf{d}_n\|} \right)' \bar{K}_n^c \left(\frac{\mathbf{u}'_n \mathbf{x}_0^{(n)}}{\|\mathbf{d}_n\|} \right) + \frac{\delta \sigma^2}{1 - \delta} \left(\text{tr } \bar{K}_n^c + (n - 2) \tilde{k}_\rho^\emptyset \right).$$

Because zero intervention is feasible,

$$0 \prec \bar{K}_n^c \preceq \begin{pmatrix} \tilde{k}_1^\emptyset & 0 \\ 0 & \tilde{k}_\rho^\emptyset \end{pmatrix}.$$

Hence $\bar{K}_n^c$ is uniformly bounded in n . Since $\mathbf{u}'_n \mathbf{x}_0^{(n)} = O(1)$ and $\|\mathbf{d}_n\| = O(1)$,

$$-\tilde{v}_n^c = O(1) + \frac{\delta \sigma^2}{1 - \delta} \left(\text{tr } \bar{K}_n^c + (n - 2) \tilde{k}_\rho^\emptyset \right).$$

Therefore, dividing each value by n ,

$$\lim_{n \rightarrow \infty} \frac{\tilde{v}_n^f}{\tilde{v}_n^m} = \frac{\tilde{k}_\rho^f}{\tilde{k}_\rho^m}, \quad \lim_{n \rightarrow \infty} \frac{\tilde{v}_n^f}{\tilde{v}_n^c} = \frac{\tilde{k}_\rho^f}{\tilde{k}_\rho^\emptyset}.$$

Since $0 < \tilde{k}_\rho^f < \tilde{k}_\rho^m < \tilde{k}_\rho^\emptyset$, the conclusion follows. □

Supplementary Appendix

S1 Additional proofs

Proof of Lemma 1: Fix $\mathbf{b} \neq \mathbf{0}$, $c > 0$, and $\delta \in (0, 1)$. We verify that the discounted linear-quadratic problem has a unique stabilizing quadratic value function. First, the no-intervention policy is feasible and yields a finite value. Under $r_t \equiv 0$,

$$\sum_{t=0}^{\infty} \delta^t \mathbf{x}_t' Q \mathbf{x}_t = \mathbf{x}' \left(\sum_{t=0}^{\infty} (\sqrt{\delta} A')^t Q (\sqrt{\delta} A)^t \right) \mathbf{x}.$$

Since A is row-stochastic, $\rho(A) \leq 1$, so $\rho(\sqrt{\delta} A) < 1$. The series therefore converges, and the value is finite for every initial state. To apply the standard discrete-time LQ theorem, write the problem in scaled variables

$$\mathbf{y}_t = (\sqrt{\delta})^t \mathbf{x}_t, \quad u_t = (\sqrt{\delta})^t r_t, \quad \text{then}$$

$$\mathbf{y}_{t+1} = \sqrt{\delta} A \mathbf{y}_t + \sqrt{\delta} \mathbf{b} u_t, \text{ and } \sum_{t=0}^{\infty} \delta^t (\mathbf{x}_t' Q \mathbf{x}_t + c r_t^2) = \sum_{t=0}^{\infty} (\mathbf{y}_t' Q \mathbf{y}_t + c u_t^2).$$

Because $\sqrt{\delta} A$ is Schur stable, the pair $(\sqrt{\delta} A, \sqrt{\delta} \mathbf{b})$ is stabilizable. Since $Q \succ 0$, the pair $(Q^{1/2}, \sqrt{\delta} A)$ is detectable. With $c > 0$, the standard discrete-time LQ theorem gives a unique stabilizing quadratic value function $v(\mathbf{x}) = -\mathbf{x}' K^* \mathbf{x}$, where $K^* \succ 0$, and a unique linear optimal policy. It remains only to record the associated equations. The Bellman equation in the original discounted variables is

$$v(\mathbf{x}) = \max_r \left\{ -\mathbf{x}' Q \mathbf{x} - c r^2 - \delta (A \mathbf{x} + \mathbf{b} r)' K^* (A \mathbf{x} + \mathbf{b} r) \right\}.$$

The first-order condition gives

$$r^*(\mathbf{x}) = -\delta (c + \delta \mathbf{b}' K^* \mathbf{b})^{-1} \mathbf{b}' K^* A \mathbf{x} \equiv -\mathbf{p}' \mathbf{x}.$$

Substituting this rule into the Bellman equation and matching quadratic terms gives

$$K^* = Q + \delta A' \left(K^* - \delta K^* \mathbf{b} (c + \delta \mathbf{b}' K^* \mathbf{b})^{-1} \mathbf{b}' K^* \right) A.$$

The stabilizing solution is unique, so this equation uniquely determines the continuation value and the optimal intervention rule. $\square$

Proof of Lemma 2: Fix $\mathbf{b} \neq \mathbf{0}$ and let $\gamma > 1$. The dynamics under exposure rule $\gamma\mathbf{b}$ and intervention level $\tilde{r}_t$ are identical to those under exposure rule $\mathbf{b}$ and intervention level $r_t = \gamma\tilde{r}_t$. Hence replacing $\mathbf{b}$ by $\gamma\mathbf{b}$ is equivalent to lowering the intervention cost from c to c/γ^2 . The value weakly increases, and strictly increases whenever the optimal intervention is nontrivial. It remains to show that intervention is strictly valuable. Let K^0 denote the no-intervention loss matrix,

$$K^0 = \sum_{t=0}^{\infty} \delta^t (A')^t Q A^t.$$

Since $Q A \mathbf{x}_0 \neq \mathbf{0}$, we have $A \mathbf{x}_0 \neq \mathbf{0}$, and therefore $K^0 A \mathbf{x}_0 \neq \mathbf{0}$. Choosing an exposure direction proportional to $-K^0 A \mathbf{x}_0$ and a sufficiently small positive intervention at date zero, followed by zero intervention thereafter, strictly improves on no intervention. Hence every optimal exposure rule induces a nontrivial intervention. Therefore, if $\|\mathbf{b}\| < 1$, scaling $\mathbf{b}$ up slightly while remaining feasible strictly improves the value. Every optimal exposure rule must satisfy $\|\mathbf{b}^*\| = 1$. $\square$

Proof of Corollary 1: Fix a value of the scalar primitive θ and a stationary pair $(\mathbf{b}^*, \mu^*)$. Let $T_{\mathbf{b}^*} = \{\mathbf{h} \in \mathbb{R}^n : \mathbf{b}^{*'} \mathbf{h} = 0\}$ be the tangent space of the unit sphere at $\mathbf{b}^*$. Let $P^* \equiv I - \mathbf{b}^* \mathbf{b}^{*'}$ denote the orthogonal projection onto $T_{\mathbf{b}^*}$. Define the constrained Hessian $H(\mathbf{b}^*, \theta) : T_{\mathbf{b}^*} \rightarrow T_{\mathbf{b}^*}$ by

$$H(\mathbf{b}^*, \theta) \mathbf{h} = P^* (v_{\mathbf{b}\mathbf{b}}(\mathbf{b}^*; \theta) - \mu^* I) \mathbf{h}, \quad \mathbf{h} \in T_{\mathbf{b}^*}.$$

Equivalently,

$$H(\mathbf{b}^*, \theta) = P^* (v_{\mathbf{b}\mathbf{b}}(\mathbf{b}^*; \theta) - \mu^* I)|_{T_{\mathbf{b}^*}}.$$

By nondegeneracy, this operator is nonsingular on $T_{\mathbf{b}^*}$. The bordered derivative of the first-order condition and the constraint is

$$\begin{pmatrix} v_{\mathbf{b}\mathbf{b}}(\mathbf{b}^*; \theta) - \mu^* I & -\mathbf{b}^* \\ \mathbf{b}^{*'} & 0 \end{pmatrix}.$$

To see that this matrix is nonsingular, suppose $(\mathbf{h}, a)$ lies in its kernel. Then

$$\begin{aligned}(v_{\mathbf{b}\mathbf{b}}(\mathbf{b}^*; \theta) - \mu^* I) \mathbf{h} - a \mathbf{b}^* &= \mathbf{0}, \\ \mathbf{b}^{*\prime} \mathbf{h} &= 0.\end{aligned}$$

The second equation implies $\mathbf{h} \in T_{\mathbf{b}^*}$. Premultiplying the first equation by P^* gives $P^* (v_{\mathbf{b}\mathbf{b}}(\mathbf{b}^*; \theta) - \mu^* I) \mathbf{h} = \mathbf{0}$, because $P^* \mathbf{b}^* = \mathbf{0}$. By the definition of H , this is $H(\mathbf{b}^*, \theta) \mathbf{h} = \mathbf{0}$. Since $H(\mathbf{b}^*, \theta)$ is nonsingular on $T_{\mathbf{b}^*}$, $\mathbf{h} = \mathbf{0}$. The first equation then gives $a \mathbf{b}^* = \mathbf{0}$, and hence $a = 0$. The bordered derivative is therefore nonsingular. The implicit function theorem implies that each stationary exposure rule is locally unique and varies smoothly with the primitives. For the local comparative static, differentiate with respect to the scalar primitive θ . The first-order condition gives

$$(v_{\mathbf{b}\mathbf{b}}(\mathbf{b}^*; \theta) - \mu^* I) \frac{d\mathbf{b}^*}{d\theta} - \frac{d\mu^*}{d\theta} \mathbf{b}^* = -v_{\mathbf{b}\theta}(\mathbf{b}^*; \theta).$$

Differentiating the constraint $\mathbf{b}^{*\prime} \mathbf{b}^* = 1$ gives

$$\mathbf{b}^{*\prime} \frac{d\mathbf{b}^*}{d\theta} = 0, \quad \text{or} \quad \frac{d\mathbf{b}^*}{d\theta} \in T_{\mathbf{b}^*}.$$

Premultiplying the differentiated first-order condition by P^* eliminates the change in the multiplier:

$$P^* (v_{\mathbf{b}\mathbf{b}}(\mathbf{b}^*; \theta) - \mu^* I) \frac{d\mathbf{b}^*}{d\theta} = -P^* v_{\mathbf{b}\theta}(\mathbf{b}^*; \theta),$$

because $P^* \mathbf{b}^* = \mathbf{0}$. Since $d\mathbf{b}^*/d\theta \in T_{\mathbf{b}^*}$, the left-hand side is, by definition, $H(\mathbf{b}^*, \theta) \frac{d\mathbf{b}^*}{d\theta}$. Therefore,

$$H(\mathbf{b}^*, \theta) \frac{d\mathbf{b}^*}{d\theta} = -P^* v_{\mathbf{b}\theta}(\mathbf{b}^*; \theta).$$

Because $H(\mathbf{b}^*, \theta)$ is nonsingular on $T_{\mathbf{b}^*}$,

$$\frac{d\mathbf{b}^*}{d\theta} = -H(\mathbf{b}^*, \theta)^{-1} P^* v_{\mathbf{b}\theta}(\mathbf{b}^*; \theta).$$

Equivalently,

$$\frac{d\mathbf{b}^*}{d\theta} = -H(\mathbf{b}^*, \theta)^{-1} (I - \mathbf{b}^* \mathbf{b}^{*\prime}) v_{\mathbf{b}\theta}(\mathbf{b}^*; \theta).$$

Nondegeneracy also implies that every stationary exposure rule is isolated. Since the stationary set is closed and contained in the compact unit sphere, it is finite. Finally, at a

local maximum the constrained Hessian is negative semidefinite. Since it is nonsingular, it is negative definite, so every local maximum is strict. $\square$

Proof of Lemma 3: Fix the orthonormal eigenbasis used in the statement. Suppose by contradiction that a strict local optimum $\tilde{\mathbf{b}}^*$ has $\tilde{b}_j^* \neq 0$. For $\varepsilon \in (0, 1)$, define a nearby exposure rule $\hat{\mathbf{b}}$ by

$$\hat{b}_k = \tilde{b}_k^* \quad \text{for } k \neq j, \quad \hat{b}_j = (1 - \varepsilon)\tilde{b}_j^*.$$

Then $\|\hat{\mathbf{b}}\| < \|\tilde{\mathbf{b}}^*\| \leq 1$, so $\hat{\mathbf{b}}$ is feasible, distinct from $\tilde{\mathbf{b}}^*$, and arbitrarily close to it.

Fix any intervention sequence $\{r_t\}$. All coordinates $k \neq j$ follow the same path under $\tilde{\mathbf{b}}^*$ and $\hat{\mathbf{b}}$. In coordinate j , the two dynamics are

$$\tilde{x}_{t+1}^j = \lambda_j \tilde{x}_t^j + \tilde{b}_j^* r_t, \quad \hat{x}_{t+1}^j = \lambda_j \hat{x}_t^j + (1 - \varepsilon)\tilde{b}_j^* r_t.$$

Since $\lambda_j \tilde{x}_0^j = 0$, we have $\hat{x}_1^j = (1 - \varepsilon)\tilde{x}_1^j$. Induction then gives

$$\hat{x}_t^j = (1 - \varepsilon)\tilde{x}_t^j \quad \text{for every } t \geq 1.$$

Thus loss in coordinate j is weakly lower at every date, and strictly lower whenever $\tilde{x}_t^j \neq 0$ for some $t \geq 1$. Losses in all other coordinates and intervention costs are unchanged.

Therefore every intervention sequence available under $\tilde{\mathbf{b}}^*$ yields a weakly higher payoff under $\hat{\mathbf{b}}$. Maximizing over intervention sequences gives $\tilde{v}(\tilde{\mathbf{x}}_0; \hat{\mathbf{b}}) \geq \tilde{v}(\tilde{\mathbf{x}}_0; \tilde{\mathbf{b}}^*)$. Because $\hat{\mathbf{b}}$ is feasible, distinct from $\tilde{\mathbf{b}}^*$, and arbitrarily close to it, $\tilde{\mathbf{b}}^*$ cannot be a strict local maximum. Hence any strict local maximum must satisfy $\tilde{b}_j^* = 0$. $\square$

Proof of Corollary 2: If $\rho = 0$ and $\bar{x}_0 = 0$, then $A(\rho)\mathbf{x}_0 = \mathbf{0}$. Zero intervention is optimal for every exposure rule, so the value is constant in $\mathbf{b}$ and there is no strict local maximum. Otherwise, $A(\rho)\mathbf{x}_0 \neq \mathbf{0}$. Any strict local maximum must have unit norm. Indeed, if its optimal intervention is nontrivial, scaling the exposure rule outward slightly implements the same opinion path at strictly lower intervention cost. If its optimal intervention is identically zero, then zero intervention remains feasible under every nearby exposure rule, so it cannot be a strict local maximum.

Fix ρ . As in the text, write the initial opinion vector as $\mathbf{x}_0 = \sqrt{n}\bar{x}_0\mathbf{u} + \mathbf{d}$, where $\mathbf{u} = \mathbf{1}/\sqrt{n}$ and $\mathbf{d} \perp \mathbf{u}$. Consider any exposure rule with $\|\mathbf{b}\| = 1$. Decompose it into its average-opinion

component and its disagreement component in a similar way. Let $\mathbf{b} = a\mathbf{u} + s\mathbf{h}$, where $\mathbf{h} \perp \mathbf{u}$, $\|\mathbf{h}\| = 1$, and $a^2 + s^2 = 1$. If $s = 0$, then $\mathbf{b} = a\mathbf{u} \in \text{span}\{\mathbf{u}, \mathbf{d}\}$, so there is nothing to prove. Suppose $s > 0$. Given the disagreement direction $\mathbf{h}$ chosen by the exposure rule, decompose initial disagreement as

$$\mathbf{d} = z\mathbf{h} + \mathbf{d}^\perp, \quad z = \mathbf{h}'\mathbf{d}, \quad \mathbf{d}^\perp \perp \mathbf{h}.$$

Then $\|\mathbf{d}^\perp\|^2 = \|\mathbf{d}\|^2 - z^2$. Since $A(\rho)\mathbf{u} = \mathbf{u}$ and $A(\rho)\mathbf{y} = \rho\mathbf{y}$ for every $\mathbf{y} \perp \mathbf{u}$, the subspace spanned by $\mathbf{u}$ and $\mathbf{h}$ is invariant. On this subspace, the controlled coordinates are $m_t = \mathbf{u}'\mathbf{x}_t$ and $y_t = \mathbf{h}'\mathbf{x}_t$, with dynamics $m_{t+1} = m_t + ar_t$ and $y_{t+1} = \rho y_t + sr_t$. All residual disagreement, orthogonal to both $\mathbf{u}$ and $\mathbf{h}$, receives no intervention and evolves autonomously with eigenvalue ρ . Thus $\mathbf{d}_{t+1}^\perp = \rho\mathbf{d}_t^\perp$.

Let $K_2(a, s)$ be the Riccati matrix for the two-dimensional controlled system, and let $k_{ij}(a, s)$ denote its ij -th entry. The residual disagreement has discounted loss coefficient

$$k_\rho = \sum_{t=0}^{\infty} \delta^t \rho^{2t} = \frac{1}{1 - \delta\rho^2}.$$

Hence the value generated by the exposure rule $\mathbf{b} = a\mathbf{u} + s\mathbf{h}$ is

$$v(\mathbf{x}_0; \mathbf{b}) = -k_{11}(a, s)n\bar{x}_0^2 - 2k_{12}(a, s)\sqrt{n}\bar{x}_0z - (k_{22}(a, s) - k_\rho)z^2 - k_\rho\|\mathbf{d}\|^2.$$

For fixed (a, s) , the only remaining choice is the disagreement direction $\mathbf{h}$, and this choice affects value only through $z = \mathbf{h}'\mathbf{d}$. Because the decision-maker can always choose zero intervention, the optimized loss from the controlled disagreement coordinate cannot exceed its uncontrolled loss. Hence $k_{22}(a, s) \leq k_\rho$, so $v(\mathbf{x}_0; \mathbf{b})$ is convex in z . If $|z| < \|\mathbf{d}\|$, then $\mathbf{h}$ can be rotated locally within the disagreement space so as to move z in either direction. Since value is convex in z , it cannot have a strict local maximum at such an interior value. Any strict local maximum must therefore satisfy $|z| = \|\mathbf{d}\|$. By equality in the Cauchy-Schwarz inequality, this holds if and only if $\mathbf{h} = \pm \frac{\mathbf{d}}{\|\mathbf{d}\|}$. Hence the disagreement component of $\mathbf{b}$ is proportional to $\mathbf{d}$, and therefore $\mathbf{b} \in \text{span}\{\mathbf{u}, \mathbf{d}\}$. $\square$

Proof of Lemma 4: Define the gap matrix $G(\mathbf{b}) \equiv K^*(\mathbf{b}) - K^f$. We first show that $G(\mathbf{b})$

satisfies the Lyapunov equation

$$G(\mathbf{b}) = \Phi(\mathbf{b}) + \delta \tilde{A}' G(\mathbf{b}) \tilde{A},$$

and then establish a series representation.

Step 1: Riccati matrix identities. Recall that for any coarse exposure rule $\mathbf{b}$, we have $\mathbf{p} = \mathbf{p}(\mathbf{b})$ and $\tilde{A}(\mathbf{b}) = A - \mathbf{b}\mathbf{p}'$. Under the optimal scalar intervention level $r_t = -\mathbf{p}'\mathbf{x}_t$, the induced vector intervention is $\mathbf{u}_t = -\mathbf{b}\mathbf{p}'\mathbf{x}_t$, and the coarse transition matrix is $\tilde{A} = A - \mathbf{b}\mathbf{p}'$. The value iteration for the coarse problem can be rewritten as

$$K^*(\mathbf{b}) = Q + c\mathbf{p}\mathbf{p}' + \delta \tilde{A}' K^*(\mathbf{b}) \tilde{A}. \quad (\text{S13})$$

The term $c\mathbf{p}\mathbf{p}'$ comes from the intervention cost

$$cr_t^2 = c(\mathbf{p}'\mathbf{x}_t)^2 = \mathbf{x}_t' c\mathbf{p}\mathbf{p}' \mathbf{x}_t.$$

Under full targeting, the optimal vector intervention is $\mathbf{u}_t = -F^f \mathbf{x}_t$, where

$$F^f = \delta(cI + \delta K^f)^{-1} K^f A.$$

The full-targeting value iteration satisfies

$$K^f = Q + c(F^f)' F^f + \delta(A - F^f)' K^f (A - F^f).$$

We use the following completion-of-squares identity. For any feedback matrix $F \in \mathbb{R}^{n \times n}$,

$$cF'F + \delta(A - F)' K^f (A - F) = K^f - Q + (F - F^f)' (cI + \delta K^f) (F - F^f). \quad (\text{S14})$$

To verify this identity, expand the right-hand side and use

$$(cI + \delta K^f) F^f = \delta K^f A, \quad (F^f)' (cI + \delta K^f) = \delta A' K^f,$$

together with the Riccati equation for K^f . Apply (S14) to the coarse feedback matrix $F^c(\mathbf{b}) = \mathbf{b}\mathbf{p}'$. Since $\|\mathbf{b}\| = 1$, we have $cF'F = c(\mathbf{b}\mathbf{p}')'(\mathbf{b}\mathbf{p}') = c\mathbf{p}\mathbf{b}'\mathbf{b}\mathbf{p}' = c\mathbf{p}\mathbf{p}'$. Hence

$$c\mathbf{p}\mathbf{p}' + \delta \tilde{A}' K^f \tilde{A} = K^f - Q + \Phi(\mathbf{b}), \quad (\text{S15})$$

where

$$\Phi(\mathbf{b}) \equiv (F^c(\mathbf{b}) - F^f)'(cI + \delta K^f)(F^c(\mathbf{b}) - F^f),$$

Since $cI + \delta K^f > 0$, we have $\Phi(\mathbf{b}) \geq 0$.

Step 2: gap Lyapunov representation. Now return to the coarse identity (S13). Add and subtract $\delta \tilde{A}' K^f \tilde{A}$:

$$K^*(\mathbf{b}) = Q + \left[c\mathbf{p}\mathbf{p}' + \delta \tilde{A}' K^f \tilde{A} \right] + \delta \tilde{A}' (K^*(\mathbf{b}) - K^f) \tilde{A}.$$

Using (S15) and canceling Q gives

$$K^*(\mathbf{b}) = K^f + \Phi(\mathbf{b}) + \delta \tilde{A}' (K^*(\mathbf{b}) - K^f) \tilde{A}. \quad (\text{S16})$$

Therefore, with $G(\mathbf{b}) = K^*(\mathbf{b}) - K^f$,

$$G(\mathbf{b}) = \Phi(\mathbf{b}) + \delta \tilde{A}' G(\mathbf{b}) \tilde{A}.$$

Step 3: Series representation. It remains to iterate (S1). Repeated substitution gives, for every $T \geq 0$,

$$G(\mathbf{b}) = \sum_{t=0}^T \delta^t \tilde{A}^{t'} \Phi(\mathbf{b}) \tilde{A}^t + \delta^{T+1} \tilde{A}^{(T+1)'} G(\mathbf{b}) \tilde{A}^{T+1}.$$

Because $\rho(\sqrt{\delta} \tilde{A}) < 1$, the final term converges to zero as $T \rightarrow \infty$. Hence

$$G(\mathbf{b}) = \sum_{t=0}^{\infty} \delta^t \tilde{A}^{t'} \Phi(\mathbf{b}) \tilde{A}^t.$$

Finally, $\Delta(\mathbf{b}) = v^f(\mathbf{x}_0) - v(\mathbf{x}_0; \mathbf{b}) = \mathbf{x}_0' (K^*(\mathbf{b}) - K^f) \mathbf{x}_0 = \mathbf{x}_0' G(\mathbf{b}) \mathbf{x}_0$ as claimed. $\square$

S2 Persistent bias and coarse targeting

Suppose opinions evolve according to

$$\mathbf{x}_{t+1} = A\mathbf{x}_t + \mathbf{b}r_t + \mathbf{w}, \quad (\text{S17})$$

where $\mathbf{w}$ is time invariant. Define the augmented state

$$\boldsymbol{\chi}_t = \begin{bmatrix} \mathbf{x}_t \\ \mathbf{w} \end{bmatrix}, \quad \bar{A} = \begin{bmatrix} A & I \\ \mathbf{0} & I \end{bmatrix}, \quad \bar{\mathbf{b}} = \begin{bmatrix} \mathbf{b} \\ \mathbf{0} \end{bmatrix}, \quad \bar{Q} = \begin{bmatrix} Q & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}.$$

Then $\boldsymbol{\chi}_{t+1} = \bar{A}\boldsymbol{\chi}_t + \bar{\mathbf{b}}r_t$, and the decision-maker's problem becomes

$$\max_{\{r_t\}_{t \geq 0}} - \sum_{t=0}^{\infty} \delta^t (\boldsymbol{\chi}_t' \bar{Q} \boldsymbol{\chi}_t + cr_t^2). \quad (\text{S18})$$

For any fixed $\mathbf{b} \neq \mathbf{0}$, this is a discounted linear-quadratic problem in the augmented state. Its value has the form $v(\boldsymbol{\chi}; \mathbf{b}) = -\boldsymbol{\chi}' \bar{K}^*(\mathbf{b}) \boldsymbol{\chi}$, where $\bar{K}^*(\mathbf{b})$ is symmetric and positive semidefinite and solves the associated Riccati equation. The optimal intervention is

$$r_t^* = -\bar{\mathbf{p}}(\mathbf{b})' \boldsymbol{\chi}_t, \quad \text{or} \quad r_t^* = -\mathbf{p}_x(\mathbf{b})' \mathbf{x}_t - \mathbf{p}_w(\mathbf{b})' \mathbf{w}.$$

Let $v(\mathbf{b}; \mathbf{x}_0, \mathbf{w})$ denote the maximal value in (S18). Since $\boldsymbol{\chi}_0 = [\mathbf{x}_0' \ \mathbf{w}']'$, we can write

$$v(\mathbf{b}; \mathbf{x}_0, \mathbf{w}) = -\boldsymbol{\chi}_0' \bar{K}^*(\mathbf{b}) \boldsymbol{\chi}_0 = - \begin{bmatrix} \mathbf{x}_0 \\ \mathbf{w} \end{bmatrix}' \bar{K}^*(\mathbf{b}) \begin{bmatrix} \mathbf{x}_0 \\ \mathbf{w} \end{bmatrix}. \quad (\text{S19})$$

Corollary S5. *The augmented problem is well behaved: an optimal exposure rule exists, and the value depends locally smoothly on $\mathbf{b}$ and the primitives. For generic primitives, every optimum has unit norm and satisfies*

$$\mu \mathbf{b} = \begin{bmatrix} I & \mathbf{0} \end{bmatrix} \bar{K}^* \tilde{A} Z_{\chi} \bar{\mathbf{p}}, \quad (\text{S20})$$

where $\tilde{A} = \bar{A} - \bar{\mathbf{b}}\bar{\mathbf{p}}'$ and $Z_{\chi} = \sum_{t=0}^{\infty} (\sqrt{\delta} \tilde{A})^t \boldsymbol{\chi}_0 \boldsymbol{\chi}_0' (\sqrt{\delta} \tilde{A})^t$.

Although $\bar{Q}$ is only psd, the same argument applies because $\rho(\sqrt{\delta} \bar{A}) < 1$. Projecting the augmented exposure gradient onto its first n coordinates gives (S20), while the usual scaling argument gives unit norm whenever intervention is strictly valuable, which holds generically. The decision-maker's tradeoff now depends not only on the endogenous opinion path, but

also on the interaction between opinions and the bias as summarized by Z_χ . Let

$$Z_\chi(\mathbf{b}) = \sum_{t=0}^{\infty} \delta^t \chi_t \chi_t' = \begin{bmatrix} \sum_{t \geq 0} \delta^t \mathbf{x}_t \mathbf{x}_t' & \sum_{t \geq 0} \delta^t \mathbf{x}_t \mathbf{w}' \\ \sum_{t \geq 0} \delta^t \mathbf{w} \mathbf{x}_t' & \frac{1}{1-\delta} \mathbf{w} \mathbf{w}' \end{bmatrix}.$$

The upper-left block records the endogenous opinion path, the lower-right block records the repeated presence of the bias, and the off-diagonal blocks record their interaction. Persistent bias therefore supplements rather than replaces the role of initial opinions.

Proposition S8. *Suppose A is symmetric and $Q = I$. Let $\Pi = I - \mathbf{u}\mathbf{u}'$, $\mathbf{d} = \Pi \mathbf{x}_0$, $\mathbf{w}_\perp = \Pi \mathbf{w}$, and define*

$$\mathcal{H}(\mathbf{x}_0, \mathbf{w}) = \text{span} \{ \mathbf{u}, A^{t+1} \mathbf{d}, A^t \mathbf{w}_\perp : t \geq 0 \}.$$

Then every strict local optimum satisfies $\mathbf{b}^ \in \mathcal{H}(\mathbf{x}_0, \mathbf{w})$.*

The subspace $\mathcal{H}(\mathbf{x}_0, \mathbf{w})$ contains three sources of payoff-relevant opinion movement: average opinion, disagreement inherited from $\mathbf{x}_0$ that survives at least one social update, and the network orbit of the disagreement repeatedly injected by $\mathbf{w}$. Exposure outside this subspace can only create additional opinion variation in directions containing neither inherited nor injected disagreement.

Proof of Proposition S8: Write $\mathcal{H} = \mathcal{H}(\mathbf{x}_0, \mathbf{w})$. Because A is symmetric, both $\mathcal{H}$ and $\mathcal{H}^\perp$ are invariant under A . Moreover, $A\mathbf{x}_0 + \mathbf{w} \in \mathcal{H}$. Thus, from date one onward, every component of opinions in $\mathcal{H}^\perp$ is generated only by intervention. Also, any strict local optimum must have unit norm. If its optimal intervention is nontrivial, scaling the exposure rule outward slightly implements the same opinion path at strictly lower intervention cost. If its optimal intervention is identically zero, every nearby exposure rule can reproduce the same payoff by also choosing zero intervention, so the optimum cannot be strict. Suppose that $\mathbf{b}$ has a nonzero component in $\mathcal{H}^\perp$. Write

$$\mathbf{b} = q\mathbf{h} + s\mathbf{k}, \quad q \geq 0, \quad s > 0, \quad q^2 + s^2 = 1,$$

where $\mathbf{h} \in \mathcal{H}$ and $\mathbf{k} \in \mathcal{H}^\perp$ are unit vectors. If $q = 0$, choose any unit vector $\mathbf{h} \in \mathcal{H}$. Choose

$q' > q$ sufficiently close to q , let

$$s' = \sqrt{1 - q'^2}, \quad \mathbf{b}' = q'\mathbf{h} + s'\mathbf{k},$$

and, for any intervention sequence $\{r_t\}$ under $\mathbf{b}$, define $r'_t = \frac{q}{q'}r_t$. The intervention generated inside $\mathcal{H}$ is unchanged: $q'\mathbf{h}r'_t = q\mathbf{h}r_t$. The intervention generated in $\mathcal{H}^\perp$ is reduced because

$$s'\mathbf{k}r'_t = \frac{s'q}{sq'} s\mathbf{k}r_t, \quad 0 \leq \frac{s'q}{sq'} < 1.$$

Since both subspaces are invariant under A , the opinion path inside $\mathcal{H}$ is unchanged, while the component in $\mathcal{H}^\perp$ and its associated loss are weakly reduced from date one onward. Intervention costs are also weakly lower: $(r'_t)^2 = \left(\frac{q}{q'}\right)^2 r_t^2 \leq r_t^2$, with strict inequality whenever $r_t \neq 0$. Hence, if intervention is nonzero at some date, $\mathbf{b}'$ yields a strictly higher payoff. If the optimal intervention under $\mathbf{b}$ is identically zero, choosing zero intervention under the nearby rule $\mathbf{b}'$ reproduces exactly the same opinion path and payoff. Hence $v(\mathbf{b}') \geq v(\mathbf{b})$, which already contradicts strict local optimality. Thus a strict local optimum cannot have a nonzero component in $\mathcal{H}^\perp$, and hence $\mathbf{b}^* \in \mathcal{H}(\mathbf{x}_0, \mathbf{w})$. $\square$

S3 Nonlinear updating and local disagreement dynamics

The effectiveness of coarse targeting depends on how disagreement propagates, rather than on the specific micro-foundation for opinion updating. For a smooth updating rule

$$\mathbf{x}_{t+1} = F(\mathbf{x}_t) + \mathbf{b}r_t,$$

consider local deviations around a no-intervention fixed point $\bar{\mathbf{x}}$, satisfying $F(\bar{\mathbf{x}}) = \bar{\mathbf{x}}$. Then

$$\Delta\mathbf{x}_{t+1} \approx J(\bar{\mathbf{x}})\Delta\mathbf{x}_t + \mathbf{b}r_t,$$

where $J(\bar{\mathbf{x}})$ is the Jacobian of F evaluated at $\bar{\mathbf{x}}$. Thus the relevant local object is the propagation matrix $J(\bar{\mathbf{x}})$. The baseline model is the special case in which this matrix is constant and equal to A . As an illustration, consider the Friedkin–Johnsen model (Friedkin and Johnsen, 1990),

$$\mathbf{x}_{t+1} = (I - \Lambda)W\mathbf{x}_t + \Lambda\mathbf{x}_0 + \mathbf{b}r_t,$$

where W is row-stochastic and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$, with $\lambda_i \in [0, 1]$. Let $\bar{\mathbf{x}}$ be the steady state absent intervention:

$$\bar{\mathbf{x}} = (I - \Lambda)W\bar{\mathbf{x}} + \Lambda\mathbf{x}_0.$$

Defining deviations from this steady state by $\Delta\mathbf{x}_t = \mathbf{x}_t - \bar{\mathbf{x}}$, we obtain

$$\Delta\mathbf{x}_{t+1} = (I - \Lambda)W\Delta\mathbf{x}_t + \mathbf{b}r_t.$$

Hence the model takes the same local form as the baseline model, with

$$A = (I - \Lambda)W.$$

Here A is substochastic when some agents are partially stubborn. If stubbornness anchors every recurrent component of the network, then $\rho(A) < 1$, so deviations decay absent intervention. Components associated with eigenvalues close to one, however, decay slowly and therefore behave like persistent disagreement for a patient decision-maker. Since Friedkin–Johnsen updating is affine, this representation is exact rather than merely local.

The same local logic applies to nonlinear models with homophily (Golub and Jackson, 2012) or persuasion bias (DeMarzo et al., 2003). Around a fixed point, the Jacobian determines which disagreement dimensions propagate and how slowly they decay. If the local propagation matrix has the island structure, the relevant dimensions are group disagreements as in the island model; if it has clustered structure, coarse targeting responds locally to slow-moving cluster disagreements. Thus the mechanism is robust within the region where the linear approximation is accurate: coarse targeting prioritizes disagreement dimensions that decay slowly, even when all disagreement is ultimately transient.

S4 Computational algorithm

S4.1 Comparison of algorithms

In general, the optimal coarse targeting weights do not admit a closed-form solution, so computation relies on numerical methods that exploit the fixed-point characterization in equation (7). This appendix describes the algorithms used to compute the optimal target vector $\mathbf{b}^*$. We consider two approaches. The first is a *nested fixed-point* algorithm directly implied by the first-order condition in (7), which evaluates the gradient efficiently by solving

the Lyapunov equation $(\sqrt{\delta}\tilde{A})Z(\sqrt{\delta}\tilde{A})' - Z + \mathbf{x}_0\mathbf{x}_0'$. Related nested fixed-point algorithms are common in dynamic discrete choice problems (Rust, 1987). The second is a class of *projected gradient methods* (PGM) for constrained nonconvex optimization, which serve as a natural benchmark and are widely used in related settings (Gao et al., 2021; Solimine and Meyer-Baese, 2022).

Both approaches require repeated evaluation of the value function and its gradient. For a given $\mathbf{b}$, this involves solving an algebraic Riccati equation for the continuation value $K^*(\mathbf{b})$ and a discrete Lyapunov equation for the associated Gramian (Vaughan, 1970; Laub, 1979).

A. Common computational primitives. Both the nested fixed-point iteration and the projected gradient methods rely on the same underlying computational primitives. For any candidate target vector $\mathbf{b}$, the continuation value $K^*(\mathbf{b})$ is obtained as the unique stabilizing solution to an algebraic Riccati equation. Conditional on $K^*(\mathbf{b})$, the induced feedback vector $\mathbf{p}(\mathbf{b})$ and transition matrix $\tilde{A}(\mathbf{b}) = A - \mathbf{b}\mathbf{p}(\mathbf{b})'$ are computed, and the discounted Gramian associated with the induced opinion dynamics is obtained by solving a discrete Lyapunov equation. Together, these objects determine the gradient $\nabla_{\mathbf{b}}v(\mathbf{b})$ appearing on the right-hand side of the first-order condition (7).¹⁴

Algorithm 3 summarizes the steps required to compute $\nabla_{\mathbf{b}}v(\mathbf{b})$ for a given $\mathbf{b}$. Both classes of algorithms discussed below repeatedly call this routine; they differ only in how the target vector $\mathbf{b}$ is updated across iterations, not in the evaluation of the underlying matrix equations.

B. Projected gradient method (benchmark). As a benchmark, we implement a projected gradient method (PGM) for the constrained maximization of the value function $v(\mathbf{b})$ subject to the norm constraint $\|\mathbf{b}\|=1$. Projected gradient methods are standard tools for constrained, nonconvex optimization problems and have been used in related targeting environments (Gao et al., 2021).

Starting from an initial exposure vector $\mathbf{b}_0$, the method updates $\mathbf{b}$ iteratively in the direction of the gradient $\nabla_{\mathbf{b}}v(\mathbf{b}_k)$, projected onto the tangent space of the constraint set. Concretely,

¹⁴The gradient formula is valid for any symmetric positive-definite payoff matrix Q . The derivation does not differentiate Q with respect to $\mathbf{b}$; Q enters through the Riccati solution $K^*(\mathbf{b})$. Thus the same algorithm applies to the population-weighted specification Q^{pop} .

each update takes the form

$$\mathbf{b}_{k+1} = \frac{\mathbf{b}_k + \eta_k(I - \mathbf{b}_k \mathbf{b}_k') \nabla_{\mathbf{b}} v(\mathbf{b}_k)}{\|\mathbf{b}_k + \eta_k(I - \mathbf{b}_k \mathbf{b}_k') \nabla_{\mathbf{b}} v(\mathbf{b}_k)\|},$$

where $(I - \mathbf{b}_k \mathbf{b}_k')$ removes the component of the gradient that points in the direction of $\mathbf{b}_k$, and η_k is a step size.

We use an adaptive step size: η_k is selected by a line search along the projected-gradient direction to maximize the value function locally. Fixed-step implementations are sensitive to the selected step size, so the reported comparison uses adaptive PGM. Algorithm 1 summarizes the procedure. Under standard regularity conditions, PGM converges to a stationary point of the constrained problem, though convergence is generally local and may depend on initialization.

C. Nested fixed-point iteration. The second approach is a nested fixed-point iteration. Rather than updating $\mathbf{b}$ along a gradient direction with an explicit step size, this method constructs a sequence $\{\mathbf{b}_k\}$ by repeatedly evaluating the fixed-point mapping implied by the optimality condition.

Starting from an initial exposure vector $\mathbf{b}_0$, each iteration updates $\mathbf{b}$ by plugging the current iterate into the right-hand side of the first-order condition and renormalizing to satisfy the norm constraint,

$$\mathbf{b}_{k+1} \propto \nabla_{\mathbf{b}} v(\mathbf{b}_k),$$

with proportionality chosen so that $\|\mathbf{b}_{k+1}\| = 1$. Algorithm 2 summarizes this update rule.

The method is *nested* because evaluation of the fixed-point mapping at $\mathbf{b}_k$ requires solving for the continuation value $K^*(\mathbf{b}_k)$ and the associated discounted Gramian, as described in Part B. In contrast to the projected gradient method, the nested fixed-point iteration does not require choosing a step size or performing a line search; the update is entirely determined by the fixed-point characterization of optimal targeting. For a given iterate $\mathbf{b}_k$, the matrix equation defining $K^*(\mathbf{b}_k)$ is solved directly using a standard Schur decomposition–based algorithm for algebraic Riccati equations (Laub, 1979). This avoids iterative solution of the inner fixed point and allows each outer iteration to be computed reliably using well-established numerical linear algebra routines.

Algorithm 1: Adaptive PGM

Data: $A, \mathbf{x}_0, \delta, c, Q$ **. Result:** Vector of optimal targetweights $\mathbf{b}^*$ $\mathbf{b}_0 \leftarrow \frac{A\mathbf{x}_0}{\|A\mathbf{x}_0\|}$ if $A\mathbf{x}_0 \neq \mathbf{0}$; otherwise $\mathbf{b}_0 \leftarrow \mathbf{e}_1 \quad k \leftarrow 0$;**while not converged do** $\mathbf{g}_k \leftarrow \nabla_{\mathbf{b}} v(\mathbf{b}_k)$; $\mathbf{p}\mathbf{g}_k \leftarrow (I - \frac{\mathbf{b}_k \mathbf{b}_k'}{\mathbf{b}_k' \mathbf{b}_k}) \mathbf{g}_k$;Define $\mathbf{f}_k(\eta) = \frac{\mathbf{b}_k + \eta \mathbf{p}\mathbf{g}_k}{\|\mathbf{b}_k + \eta \mathbf{p}\mathbf{g}_k\|}$; $\eta_k^* \leftarrow \arg \max_{\eta} v(\mathbf{f}_k(\eta))$; $\mathbf{b}_{k+1} \leftarrow \mathbf{f}_k(\eta_k^*)$; $k \leftarrow k + 1$;**end****return** $\mathbf{b}_k$;

Algorithm 2: Nested fixed-point

Data: $A, \mathbf{x}_0, \delta, c, Q$ **. Result:** Vector of optimal targetweights $\mathbf{b}^*$ $\mathbf{b}_0 \leftarrow \frac{A\mathbf{x}_0}{\|A\mathbf{x}_0\|}$ if $A\mathbf{x}_0 \neq \mathbf{0}$; otherwise $\mathbf{b}_0 \leftarrow \mathbf{e}_1 \quad k \leftarrow 0$;**while not converged do** $\mathbf{b}_{k+1} \leftarrow \nabla_{\mathbf{b}} v(\mathbf{b}_k)$; $\mathbf{b}_{k+1} \leftarrow \mathbf{b}_{k+1} / \|\mathbf{b}_{k+1}\|$; $k \leftarrow k + 1$;**end****return** $\mathbf{b}_k$;

Algorithm 3: Compute gradient $\nabla_{\mathbf{b}} v(\mathbf{b})$

Data: $\mathbf{b}, A, \delta, c, Q$ **Result:** Gradient $\nabla_{\mathbf{b}} v(\mathbf{b})$ Solve $K^* = Q + \delta A' K^* (I + \frac{\delta}{c} \mathbf{b} \mathbf{b}' K^*)^{-1} A$ for K^* ; $\mathbf{p}' \leftarrow (c + \delta \mathbf{b}' K^* \mathbf{b})^{-1} \delta \mathbf{b}' K^* A$; $\tilde{A} \leftarrow A - \mathbf{b} \mathbf{p}'$;Solve $\sqrt{\delta} \tilde{A} Z \sqrt{\delta} \tilde{A}' - Z = -\mathbf{x}_0 \mathbf{x}_0'$ for Z ;**return** $2\delta K^* \tilde{A} Z \mathbf{p}$;

D. Practical implementation details. Both algorithms require an initial exposure vector and a stopping criterion. Throughout the numerical analysis, we initialize the algorithms at $\mathbf{b}_0 = \frac{A\mathbf{x}_0}{\|A\mathbf{x}_0\|}$, a natural starting point aligned with both opinions and the network structure. We terminate the algorithms when successive iterates satisfy $\|\mathbf{b}_{k+1} - \mathbf{b}_k\| < \varepsilon$, or when the norm of the projected gradient falls below a tolerance level, with ε set to 10^{-6} in all reported simulations. All algorithms are also subject to a maximum iteration cap to ensure termination.

In practice, the nested fixed-point iteration converges reliably for a wide range of networks

and preference parameters. In some cases, however, the fixed-point mapping may be unstable or fail to converge. When this occurs, we switch to the adaptive projected gradient method, which converges more robustly albeit at higher computational cost. Importantly, when both methods converge, they yield identical solutions for the optimal exposure vector $\mathbf{b}^*$.

E. Simulation design. To evaluate the numerical performance of the algorithms described above, we conduct simulations on synthetic networks of varying size and structure. The goal of these experiments is not to assess economic outcomes, but to compare convergence behavior and computational cost across methods under controlled conditions.

We use two simulation panels. The algorithm-comparison panel uses $n \in \{2, 4, \dots, 32\}$ and 25 independent networks at each size. A separate scalability panel uses $n \in \{25, 50, 100, 150, 200, 300, 400, 600\}$ and 10 independent networks at each size. Adaptive PGM is run through $n = 200$, while the fixed-point method is run for every listed size. Each network is drawn from a uniform Dirichlet ensemble: each row of A is sampled independently from a Dirichlet distribution on the probability simplex, ensuring that A is row-stochastic and dense. This design produces asymmetric networks with heterogeneous influence patterns, similar in spirit to the empirical Facebook network used in the application.

For each network, we draw an initial opinion vector $\mathbf{x}_0$ as follows:

$$\mu_s \sim \mathcal{N}(0, 5), \quad \sigma_s \sim \text{Uniform}(0, 5), \quad \mathbf{x}_0 \sim \mathcal{N}(\mu_s \mathbf{1}, \sigma_s^2 I).$$

This construction generates both low- and high-dispersion initial conditions, allowing us to assess algorithm performance across a range of disagreement profiles.

F. Results. Figure 5 summarizes the comparison panel. It reports wall time and the number of iterations required for convergence as a function of network size, averaged across simulation draws. In the comparison panel, nested fixed-point averages 4.13 outer iterations and 0.018 seconds per iteration¹⁵. Adaptive PGM averages fewer iterations (2.26) but 0.49s per instance, due to the expensive line search operation. The scalability panel shows the same efficiency ranking. The fixed-point method averages approximately four iterations through $n = 600$ and 5.1 iterations at $n = 1000$. At $n = 200$, it converges in 34 seconds on average, compared with 1,032 seconds for the adaptive PGM. Each iteration of both methods involves solving matrix equations with $\mathcal{O}(n^3)$ complexity. Adaptive PGM

¹⁵Simulations were run on a 2024 Intel i7 evo, with 32GB RAM.

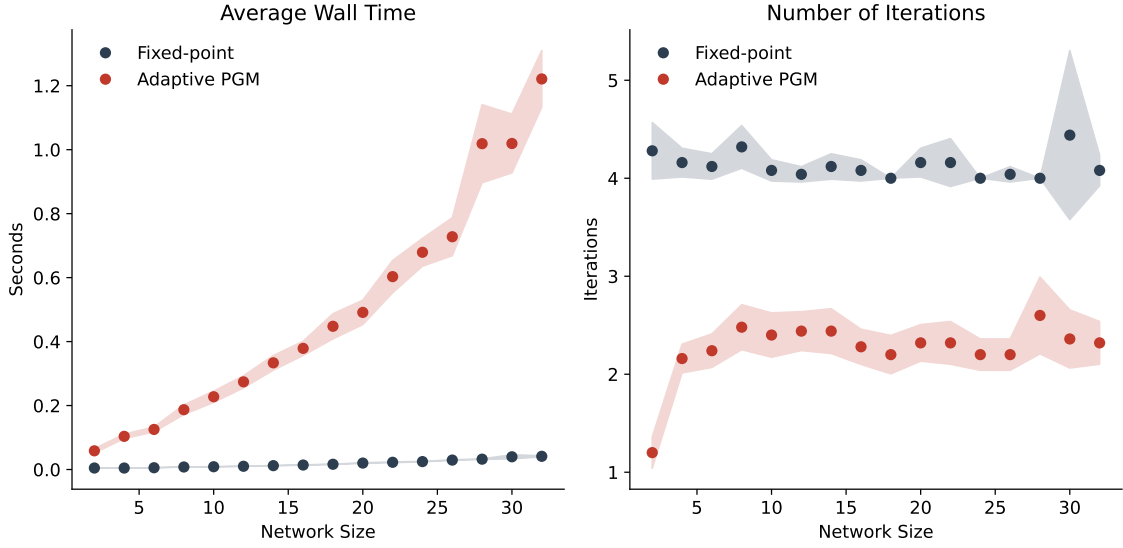


Figure 5: Convergence speed in the algorithm-comparison panel ($n \leq 32$)

additionally incurs repeated value-function evaluations during line search, leading to a per-iteration cost of order $\mathcal{O}(\log(1/\epsilon) n^3)$, where ϵ denotes the line-search tolerance.¹⁶ This difference reflects the fact that the fixed-point update directly implements the first-order condition, whereas gradient-based methods perform a sequence of local ascent and line-search steps.

In cases where the fixed-point mapping is unstable, the adaptive projected gradient method reliably converges to the same solution. In practice whenever both methods converge, they return identical optimal target vectors. The comparison therefore highlights a trade-off between speed and robustness rather than differences in optimality.

S5 Data appendix

S5.1 Proof and construction details for Section 6.1

The model presented throughout the paper takes the opinion propagation matrix A as a primitive, but the construction of the empirical object from the available data is not trivial. We develop a construction that is defensible under a natural microfoundation, connecting

¹⁶Parallel or approximate solvers (Laine and Tomlin, 2019) may reduce this gap for large or sparse networks. Each reported instance is solved serially; independent simulation draws may be distributed across processors.

individual-level social interactions to state-level opinion dynamics. The argument has two steps: first, we derive the state-level form of A from an individual population-level model. Second, we show that the unknown scale of the SCI matrix is cancelled through row-normalization, so that A is identified directly from the observable aggregate data.

We assume that individual initial opinions are distributed around the state-level average

$$y_{0,i} = x_{0,s(i)} + \epsilon_i,$$

where the shocks ϵ_i are independent across individuals and social ties, mean zero, and have finite variance σ_ϵ^2 . Opinions follow a DeGroot updating process at the individual level: $y_{t+1,i} = \sum_j T_{ij}^{\text{ind}} y_{t,j}$, where $T_{ij}^{\text{ind}} = A_{ij}^{\text{ind}} / d_i^{\text{ind}}$ is the row-stochastic normalization of the individual adjacency matrix.

Social ties form according to a state-level stochastic block model: conditional on state memberships, $A_{ij}^{\text{ind}} \sim \text{Bernoulli}(p_{s(i),s(j)})$ independently across pairs. The cross-state probability matrix $\mathbf{P} = [p_{ss'}]$ is the structural object that summarizes the model. The next result gives a fixed-date large-population justification for the state-level recursion.

Proposition S9. *Fix the number of states. Let $N = \sum_s N_s$, $N_{\min} = \min_s N_s$, and suppose $N_{\min}/N$ is bounded away from zero. Suppose also that there is a constant $\underline{\mu} > 0$ such that every expected individual degree satisfies $\mu_s \geq \underline{\mu}N$. Let*

$$x_{t,s}^{(N)} = \frac{1}{N_s} \sum_{i:s(i)=s} y_{t,i}$$

denote the realized state-average opinion. Define

$$\hat{A}_{ss'}^{(N)} = \frac{p_{ss'} N_{s'}}{\mu_s}, \quad \mu_s = \sum_{s''=1}^n p_{ss''} N_{s''}, \quad (\text{S21})$$

and let $\bar{\mathbf{x}}_0 = (x_{0,1}, \dots, x_{0,n})'$. For every fixed date t ,

$$\max_s \left| x_{t,s}^{(N)} - \left((\hat{A}^{(N)})^t \bar{\mathbf{x}}_0 \right)_s \right| \xrightarrow{p} 0 \quad \text{as } N_{\min} \rightarrow \infty.$$

Moreover, $\hat{A}^{(N)}$ is row-stochastic.

Proof of Proposition S9: Let $R^{(N)}$ be the state-averaging matrix, let $P^{(N)} = \mathbb{E}[A^{\text{ind}}]$, and define $\bar{D}^{(N)} = \text{diag}(\mu_{s(1)}, \dots, \mu_{s(N)})$ and $\bar{T}^{(N)} = (\bar{D}^{(N)})^{-1}P^{(N)}$. Thus

$$\bar{T}_{ij}^{(N)} = \frac{p_{ss'}}{\mu_s} \quad \text{when } s(i) = s, s(j) = s', \quad R^{(N)}\bar{T}^{(N)} = \hat{A}^{(N)}R^{(N)}.$$

Applying the rectangular Matrix Bernstein inequality to $A^{\text{ind}} - P^{(N)}$ and the scalar Bernstein inequality to the individual degrees gives

$$\|T^{\text{ind}} - \bar{T}^{(N)}\|_2 = O_p\left(\sqrt{\frac{\log N}{N}}\right); \quad (\text{S22})$$

see Tropp (2012, Theorem 1.6) and the related random-graph concentration results in Chung and Radcliffe (2011); Le et al. (2017). The entries of $\bar{T}^{(N)}$ are at most $(\underline{\mu}N)^{-1}$, so $\|\bar{T}^{(N)}\|_2 \leq \underline{\mu}^{-1}$. Equation (S22) then also gives $\|T^{\text{ind}}\|_2 = O_p(1)$.

For the fixed date t , the telescoping identity

$$(T^{\text{ind}})^t - (\bar{T}^{(N)})^t = \sum_{\ell=0}^{t-1} (T^{\text{ind}})^\ell (T^{\text{ind}} - \bar{T}^{(N)}) (\bar{T}^{(N)})^{t-1-\ell}.$$

It follows that

$$\|(T^{\text{ind}})^t - (\bar{T}^{(N)})^t\|_2 = O_p\left(\sqrt{\frac{\log N}{N}}\right) = o_p(1).$$

Writing the initial individual-opinion vector as $\mathbf{y}_0 = L^{(N)}\bar{\mathbf{x}}_0 + \boldsymbol{\epsilon}$, where $L^{(N)}$ replicates each state mean across its residents, gives

$$\mathbf{x}_t^{(N)} - (\hat{A}^{(N)})^t \bar{\mathbf{x}}_0 = R^{(N)}[(T^{\text{ind}})^t - (\bar{T}^{(N)})^t] \mathbf{y}_0 + (\hat{A}^{(N)})^t R^{(N)} \boldsymbol{\epsilon}.$$

Now $\|R^{(N)}\|_2 = N_{\min}^{-1/2}$ and $\|\mathbf{y}_0\|_2 = O_p(\sqrt{N})$, so the first term is $o_p(1)$. The s th component of $R^{(N)}\boldsymbol{\epsilon}$ has variance σ_ϵ^2/N_s , and hence

$$\|R^{(N)}\boldsymbol{\epsilon}\|_2 = O_p(N_{\min}^{-1/2}) = o_p(1).$$

Since $(\hat{A}^{(N)})^t$ is row-stochastic and n is fixed, its operator norm is bounded, so the second

term is also $o_p(1)$. Finally,

$$\sum_{s'} \hat{A}_{ss'}^{(N)} = \frac{\sum_{s'} p_{ss'} N_{s'}}{\mu_s} = 1,$$

which proves row-stochasticity. $\square$

Remark 1. Proposition S9 is an asymptotic, fixed-date result and does not provide a finite-sample error constant. Large state populations make state-average idiosyncratic-opinion noise small, but the theorem alone does not quantify the exact magnitude of the approximation error of the aggregated propagation matrix in a finite sample.

Remark 2. The construction does not require $\hat{A}$ and $\mathbf{x}_0$ to be independent. For example, states may have latent characteristics $\mathbf{v}_s$ that jointly determine average opinions, $x_{0,s} = f_0(\mathbf{v}_s) + \epsilon_s$, and link propensities, $p_{ss'} = W(\mathbf{v}_s, \mathbf{v}_{s'})$. Proposition S9 conditions on these state characteristics, so correlation between initial opinions and network connectivity does not alter the aggregation argument.

Proposition S9 shows that the aggregate opinion dynamics can be approximated by the state-level cross-state probability matrix $\mathbf{P} = [p_{ss'}]$, the core object of relevance to the model. Due to privacy restrictions, however, the SCI provides a signal proportional to this link probability. The following result shows that the unknown scale does not affect the propagation matrix.

Lemma S5. *Suppose $SCI_{ss'} = \kappa \cdot p_{ss'}$ for an unknown scalar $\kappa > 0$ that is common across state pairs. Then $\hat{A}^{(N)}$ defined in (S21) satisfies*

$$\hat{A}_{ss'}^{(N)} = \frac{SCI_{ss'} N_{s'}}{\sum_{s''=1}^n SCI_{ss''} N_{s''}},$$

so κ cancels identically in the row-normalization. Therefore $\hat{A}^{(N)}$ is point-identified from the observed SCI and state populations, regardless of the scale factor.

Proof of Lemma S5: Since $p_{ss'} = SCI_{ss'}/\kappa$, substituting into (S21) gives

$$\hat{A}_{ss'}^{(N)} = \frac{(SCI_{ss'}/\kappa) N_{s'}}{\sum_{s''} (SCI_{ss''}/\kappa) N_{s''}} = \frac{SCI_{ss'} N_{s'}}{\sum_{s''} SCI_{ss''} N_{s''}}. \quad (\text{S23})$$

The common factor $1/\kappa$ cancels. □

Taken together, Proposition S9 and Lemma S5 provide a fixed-date large-population justification for using the row-normalized, population-weighted SCI matrix as the empirical counterpart of the model primitive A .

Following Bailey et al. (2021), we aggregate county-level SCI to states according to

$$\text{SCI}_{ab} = \sum_{i \in C(a), j \in C(b)} \frac{\text{Pop}_i}{\text{Pop}_a} \frac{\text{Pop}_j}{\text{Pop}_b} \text{SCI}_{ij}.$$

This population-weighted average preserves proportionality to the state-pair link probability with the same scale κ . We then multiply each destination column by its state population and row-normalize as in Lemma S5. The resulting matrix is generally asymmetric because destination populations and row-normalization denominators differ across states, even though the underlying SCI is symmetric. Opinions are population-share-weighted county “consensus oppose” averages from the 2023 edition of the Yale Climate Opinion Maps (Howe et al., 2015; Marlon et al., 2022); county population shares are from Census estimates.

S5.2 Computation and robustness

We compute the coarse rule using Algorithm 2, the nested fixed-point iteration implied by (7), from several informed initializations. Each step solves an algebraic Riccati equation and a discrete Lyapunov equation. We retain the highest-value candidate and compare it with every reported fixed heuristic, so the reported coarse rule weakly dominates those included benchmarks. Full targeting uses the unrestricted vector intervention $\mathbf{u}_t$ defined in Section 2. For any payoff matrix Q , the no-intervention loss matrix solves $K^\varnothing = Q + \delta A' K^\varnothing A$. The primary specification uses $Q = Q^{\text{pop}}$. Authority and hub centrality are the leading right and left singular vectors of $\hat{A}$.

Table 4 reports the population-weighted specification on the (δ, c) grid. The last column uses a separate raw-value normalization, $|v^c - v^f|/|v^f|$, presenting the value gap as a proportion of full targeting value. Through this lens, the coarse rule stays within 0.71% of full targeting under all tested parameter values. The capture columns show that the computed coarse rule dominates every reported fixed heuristic at every cell, while each individual heuristic has a regime where it falls short.

Table 4: Population-weighted capture $(v(\mathbf{b}) - v^\emptyset)/(v^f - v^\emptyset)$ across (δ, c) ($Q = Q^{\text{pop}}$). Full targeting is the 100% benchmark; the last column is the coarse-to-full gap $|v^c - v^f|/|v^f|$.

δ	c	Coarse	Authority	Opinion	Hub	coarse/full gap
0.80	1	99.82%	95.39%	96.14%	97.60%	0.45%
0.80	10	99.87%	98.49%	81.64%	94.64%	0.13%
0.80	100	99.99%	95.80%	56.79%	82.33%	0.00%
0.90	1	99.90%	97.23%	98.30%	98.63%	0.58%
0.90	10	99.91%	99.07%	91.10%	97.52%	0.23%
0.90	100	99.99%	97.76%	71.96%	89.84%	0.01%
0.95	1	99.95%	98.49%	99.20%	99.26%	0.65%
0.95	10	99.95%	99.48%	95.67%	98.80%	0.30%
0.95	100	100.0%	98.88%	84.48%	94.77%	0.01%
0.99	1	99.99%	99.68%	99.85%	99.84%	0.71%
0.99	10	99.99%	99.89%	99.16%	99.77%	0.36%
0.99	100	100.0%	99.78%	96.73%	98.96%	0.01%

Table 5 reports population-weighted capture under recurrent shocks of size σ at $\delta = 0.9$ and $c = 10$. Opinions are shares, and the observed cross-state standard deviation of initial disagreement is about 0.036; thus $\sigma = 0.1$ is already almost three times the empirical dispersion. For each σ , we recompute the coarse direction using the expected discounted opinion-path matrix, which adds the shock covariance to the deterministic path. Both fixed-direction rules lose out to full targeting as σ grows.

Table 5: Population-weighted capture under recurrent shocks of size σ (with $\delta = 0.9$, $c = 10$, $Q = Q^{\text{pop}}$). Full targeting is the 100% benchmark at every σ .

σ	0	0.1	0.2	0.3	0.5	1.0
Coarse*	99.9%	97.2%	91.3%	85.6%	78.2%	72.4%
Authority ($\mathbf{v}_1$)	99.1%	96.5%	90.8%	85.1%	77.7%	71.6%

S5.3 Effective dimension of the opinion path

Table 6 reports the concentration of the population-weighted realized path $Z(\mathbf{b}^*)$ across the parameter grid, underlying the effective-dimension discussion in Section 6.3. The effective path dimension stays near one throughout, so the path remains low-dimensional regardless of (δ, c) . Let $\bar{\phi} = \text{tr}(\Phi)/51$. The penultimate column reports the intervention error on the

dominant unit eigenvector $v_{\max}$ of Z relative to this mean, $v'_{\max} \Phi v_{\max} / \bar{\phi}$. The final column reports the realized coarse-to-full loss as a share of attainable gains.

Table 6: Concentration of the population-weighted realized opinion path $Z(\mathbf{b}^*)$, 51-state network ($Q = Q^{\text{pop}}$). “Lead share” is the largest eigenvalue of Z over its trace; $d_{\text{eff}} = (\sum_k \omega_k)^2 / \sum_k \omega_k^2$; $v'_{\max} \Phi v_{\max} / \bar{\phi}$ is the intervention error on Z ’s leading direction relative to its mean.

δ	c	Z lead share	d_{eff}	$v'_{\max} \Phi v_{\max} / \bar{\phi}$	Δ / gain
0.80	1	96.72%	1.07	9.0×10^{-4}	0.185%
0.80	10	99.02%	1.02	1.8×10^{-4}	0.127%
0.80	100	99.79%	1.00	6.5×10^{-7}	0.007%
0.90	1	96.14%	1.08	1.3×10^{-3}	0.101%
0.90	10	98.59%	1.03	3.2×10^{-4}	0.087%
0.90	100	99.75%	1.01	2.0×10^{-6}	0.007%
0.95	1	95.86%	1.09	1.5×10^{-3}	0.052%
0.95	10	98.34%	1.03	4.1×10^{-4}	0.050%
0.95	100	99.69%	1.01	4.7×10^{-6}	0.005%
0.99	1	95.63%	1.09	1.7×10^{-3}	0.011%
0.99	10	98.11%	1.04	5.1×10^{-4}	0.011%
0.99	100	99.61%	1.01	1.0×10^{-5}	0.001%